U.S.–China Trade War and Spillover Effects to Bystander Countries

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Abstract

The U.S.–China trade war in 2018-19 was bilateral in nature, but its effects extended to countries linked to both economies through global supply chains. This paper studies how the trade war affected Korea, a bystander country deeply integrated with China and the United States. While existing evidence on bystander countries emphasizes rerouting and trade diversion, Korea imported less from China after the trade war. We develop a simple analytical model of a global supply chain linking China, Korea, and the United States. The model shows that U.S. tariffs can reduce Korean imports from China when Chinese production exhibits increasing returns to scale: lower Chinese sales to the U.S. contract production scale, raise Chinese marginal costs, and increase the prices Korean firms pay for Chinese goods. The model also allows Chinese producers to reroute goods through Korea, generating heterogeneous effects across sectors depending on rerouting costs. Using transaction-level Korean customs data, we find that a 10 percentage-point increase in U.S. tariffs on Chinese products reduced Korean import quantities from China by about 24 percent and raised Korean import prices by about 2.5 percent. The decline is concentrated in industries exhibiting increasing returns to scale. However, in industries with high pre-shock U.S.-export exposure, Korean imports from China and exports to the U.S. both rise, consistent with rerouting through Korea. These results show that bilateral tariffs can generate both decoupling from China and indirect China-U.S. trade through third countries.

JEL Classification Codes: F13, F14, F62

Keywords: U.S.–China trade war, tariffs, bystander countries, third-country effects, decoupling, trade rerouting, global value chains

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1 Introduction

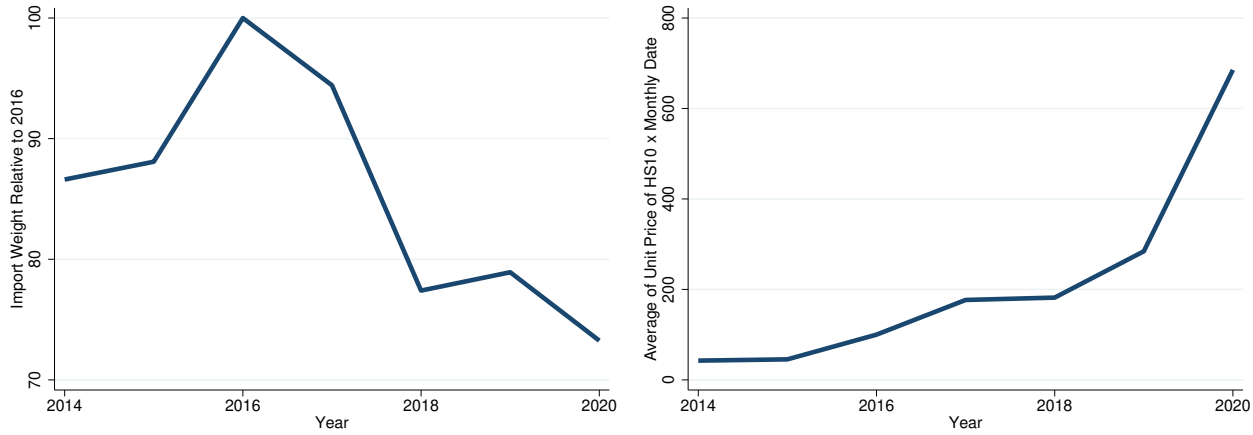
The U.S.–China trade war of 2018-19 marked one of the largest reversals in U.S. trade policy since the 1930s. Beginning in July 2018, the U.S. imposed several waves of tariffs on Chinese products, and China responded with retaliatory tariffs on U.S. exports. By late 2019, U.S. tariffs covered roughly $350 billion of imports from China, while China's retaliatory tariffs covered about $100 billion of exports from the U.S. (Fajgelbaum and Khandelwal, 2022; Bown, 2021b). Although the tariffs were imposed bilaterally, firms in other countries were exposed to the shock through their trade linkages with both economies.

The existing literature on the effects of the U.S.–China trade war on bystander countries emphasizes the rerouting of Chinese products to the U.S.: as Chinese exporters faced higher tariffs in the U.S. market, they could ship through third countries to reach the U.S. indirectly (Iyoha et al., 2024; Utar et al., 2025a). More broadly, tariff shocks can also generate trade diversion, as affected exporters redirect sales to alternative destinations (Benguria and Saffie, 2024; Jiang et al., 2023). In either case, third countries' imports from China would tend to increase.

In contrast, Korea imported less from China after the U.S.–China trade war. Figure 1 depicts the aggregate quantity and average unit value of Korean imports from China. The figure shows that import quantities from China fell after the trade war, while import prices from China increased. Motivated by this aggregate pattern, we ask two questions. First, why did Korea decouple from China in its imports when prior work on bystander countries points toward diversion and rerouting? Second, does the aggregate decline mask opposing responses across sectors?

To answer these questions, we begin by developing a simple analytical model of a global supply chain linking China, Korea, and the United States. The model highlights a new channel through which U.S. tariffs on China can reduce Korean imports from China. The key ingredient is increasing returns to scale in Chinese production. When U.S. tariffs reduce Chinese sales to the U.S. market, Chinese production scale contracts. This con-

FIGURE 1. Aggregate Korean Import Quantity and Price from China



Notes: The left panel plots aggregate Korean import quantity from China, using weight in kilograms to proxy quantity. The right panel plots the average Korean import price from China, using unit value to proxy price, defined as transaction value divided by weight. Both series cover 2014 – 2020 and are normalized to 2016. The price series reports the simple average of HS10×month unit values for imports from China in each year.

traction raises Chinese marginal costs and, in turn, the prices at which Chinese goods are supplied to Korea. Korean firms then substitute away from Chinese inputs, leading Korean imports from China to fall even though Korea is not directly targeted by the tariff.

The model also explains why this aggregate decline can mask different responses across sectors by allowing Chinese producers to choose between direct shipment to the U.S. and rerouting through Korea. Chinese producers can avoid U.S. tariffs by shipping goods through Korea, but their ability to do so depends on firm-specific rerouting costs. We introduce a parameter that governs the industry-level distribution of rerouting costs and map this parameter to Korean industries' pre-shock U.S.-export exposure in the data. When this exposure is high, rerouting through Korea is less costly, so the increase in rerouted Chinese goods can offset, or even reverse, the decline in Korean demand for Chinese goods.

Solving the industry equilibrium analytically, we derive four sets of predictions. First, U.S. tariffs on China reduce Korean imports, with the effect pronounced in industries exhibiting increasing returns to scale. Second, U.S. tariffs on China raise the prices at which

Chinese goods are supplied to Korea. Third, the decline in Korean imports from China can be weaker, or may reverse, in sectors with greater pre-shock U.S.-export exposure. Fourth, the same sectors with greater pre-shock U.S.-export exposure should also experience an increase in Korean exports to the U.S. if the import increase reflects rerouting rather than stronger Korean demand for Chinese goods.

We test these predictions using transaction-level Korean customs data covering the universe of Korean import and export transactions which we aggregate to the HS10 product-country-year level. Each observation in the panel is at the HS10 product-origin/destination country-year level. Each observation includes the value and weight, which determine the unit value per kilogram. Following the convention in the literature, we use the weight in kilograms as our proxy for trade quantity and the unit value as our proxy for the price. We combine these panel data with HS6 product-level U.S. tariffs on Chinese goods and China's retaliatory tariffs on U.S. goods during the 2018-2019 trade war.

The empirical analysis first documents that Korean imports from China declined more sharply in products exposed to larger U.S. tariff increases. A 10 percentage-point increase in the U.S. tariff on a Chinese product is associated with roughly a 24 percent decline in Korean import quantity of that product from China after 2017. We then examine the mechanism behind this decline. As implied by the model, the decline is concentrated in industries where Chinese production exhibits increasing returns to scale. We further show that Korean import prices of Chinese goods increased in products exposed to larger U.S. tariff increases. A 10 percentage-point increase in the U.S. tariff raises Korean import prices from China by about 2.5 percent. The combination of falling quantities and rising prices points to a negative supply shock propagating from China, rather than a simple diversion of Chinese goods toward Korea. We further show that the decline in Korean imports from China is concentrated in industries where Chinese production exhibits increasing returns to scale, as implied by the model.

Finally, we investigate whether the aggregate decline masks rerouting in some indus-

tries. The model predicts that rerouting should be stronger in industries where Chinese producers face lower rerouting costs. We capture this variation with a parameter governing the distribution of rerouting costs in the model and map it to Korean industries' pre-shock U.S.-export exposure in the data. This exposure reflects the trading relationships and distribution networks that firms in an industry have established in the U.S. market. We find that Korean imports from China decline in products with low U.S.-export exposure, but increase in products with high U.S.-export exposure. In the same products with high U.S.-export exposure, Korean exports to the U.S. also rise with the U.S. tariff on China. These results suggest that Korea decoupled from China on average, while also serving as a rerouting channel in industries already closely connected to the U.S. market, consistent with evidence that the trade war induced Chinese goods to move through bystander countries (Utar et al., 2025b; Iyoha et al., 2024).

Related Literature Our paper contributes to three strands of the literature. First, it relates to the large literature studying the economic consequences of the U.S.–China trade war. Existing work has documented substantial effects of the 2018–2019 tariffs on bilateral trade, prices, and economic activity in the two directly affected economies. U.S. tariffs were passed through substantially to import prices (Amiti et al. (2019, 2020); Fajgelbaum et al. (2020)), which in turn affected retailers' margins (Cavallo et al. (2021)) and consumer welfare, while also raising producer prices and reducing employment in the U.S. manufacturing sector through higher input costs (Flaaen and Pierce (2024)). On the Chinese side, firms responded by reducing exports to the United States and in some cases, redirecting shipments toward alternative destinations (Jiang et al. (2023); Jiao et al. (2024)). While this literature primarily studies the consequences of these adjustments within the United States and China, we study how Chinese producers' responses themselves become a source of spillovers to a third country. In particular, we show that tariff-induced contractions in Chinese production raise the prices faced by Korean importers and affect Korean firms' sourcing decisions and export performance.

Second, our paper contributes to the growing literature on the effects of bilateral trade conflicts on bystander countries. Fajgelbaum et al. (2024) show that the U.S.–China trade war generated substantial global trade reallocation, creating both winners and losers among countries not directly involved in the dispute. Subsequent work documents trade-diversion and local economic effects in Brazil, Mexico, Vietnam (Cavalcanti et al. (2026); Utar et al. (2025a); Mayr-Dorn et al. (2026); Iyoha et al. (2024)). Conteduca et al. (2026) further show that the effects on third-country firms depend on substitution and complementarity patterns in demand and on external economies of scale in the bystander economy. Relatedly, Ma et al. (2026) show that lower Chinese exports to the United States reduced China's demand for upstream inputs from Japan, Korea, and Taiwan. We contribute to this literature by developing a simple analytical framework that incorporates both tariff-induced cost increases among Chinese producers and rerouting through third countries, yielding testable predictions for third-country imports and exports. We further show that the relative importance of these channels can vary substantially across industries even within the same third country.

Third, our paper relates to the literature on trade policy and the reorganization of global value chains. Recent studies show that tariffs induce firms to switch suppliers, relocate production, and reorganize cross-border production networks, including through upstream and downstream linkages across production networks (Flaaen et al. (2020); Ma et al. (2026); Grossman et al. (2024); Freund et al. (2024); Garred and Yuan (2025); Iyoha et al. (2024); Benguria and Saffie (2024); Shen (2026)). A recurring finding in this literature is that a decline in direct U.S. imports from China need not imply an equivalent decline in dependence on Chinese production. In particular, emerging suppliers that gain U.S. market share often remain closely integrated with China and continue to source intermediate inputs from Chinese producers (Freund et al. (2024); Garred and Yuan (2025)). Relatedly, Antràs et al. (2026) analyze how redirected Chinese intermediate inputs can lower costs for downstream producers in third countries. We add to this literature by

highlighting a distinct margin of global value-chain adjustment beyond rerouting: U.S. tariffs raise the relative price of Chinese inputs, inducing Korean firms to reduce their reliance on Chinese suppliers. More broadly, our results show that bilateral tariffs can reshape third-country links with the targeted economy in opposite directions, strengthening them through rerouting or trade diversion in some industries while weakening input-sourcing links in others.

The rest of the paper proceeds as follows. Section 2 introduces the model environment and derives its analytical predictions. Section 3 tests these predictions using Korean customs data. Section 4 concludes.

2 Model

We present a simple model of a global supply chain composed of three countries – China (C), Korea (K), and the U.S. (U) – to study the effects of U.S. tariffs on Chinese goods on Korean imports from China and exports to the U.S. Using the model, we develop testable hypotheses about the effects of the U.S. tariffs on China, and we empirically verify them using Korean customs data. We construct a static partial equilibrium of an industry that operates in the three economies. We impose a simple structure on cross-border trade. Chinese firms sell differentiated varieties in both Korea and the United States. In Korea, Chinese varieties are purchased by consumers as final goods and by Korean firms as intermediate inputs. Korean firms combine Chinese inputs with domestic labor to produce differentiated varieties that they sell in the U.S. Chinese firms can serve the U.S. market either directly or by rerouting their goods through Korean intermediaries. For analytical tractability, we assume that Korea is a small economy relative to the U.S. and China, so Korean firms' pricing and production choices do not affect the industry output and prices in the other two countries.

2.1 Demand

We consider representative consumers in destination $d \in \{K, U\}$ that consume a composite of Chinese varieties in industry i, denoted by Q_{id}. We assume that the demand for the composite goods is isoelastic, i.e.,

$$Q_{id} = P_{id}^{-\sigma}, \tag{1}$$

where P_{id} is the industry price index and $\sigma > 0$ is the price elasticity of aggregate industry demand. For a given level of Q_{id}, the consumer minimizes expenditure across varieties, which are aggregated as

$$Q_{id} = \left(\int_{j \in \Omega_{id}^C} \left(q_{id}^C(j) \right)^{\frac{\eta-1}{\eta}} \, dj \right)^{\frac{\eta}{\eta-1}}. \tag{2}$$

where η is the elasticity of substitution across varieties, Ω_{id}^C is the set of Chinese varieties available in destination d, and $q_{id}^C(j)$ is demand for Chinese variety j in that destination. Expenditure minimization over the CES aggregate Q_{id} yields

$$q_{id}^C(j) = \left(p_{id}^C(j) \right)^{-\eta} P_{id}^{\eta-\sigma}. \tag{3}$$

where $p_{id}^C(j)$ is the price of the variety j. The industry price index P_{id} satisfies

$$P_{id}^{1-\eta} = \int_{j \in \Omega_{id}^C} \left(p_{id}^C(j) \right)^{1-\eta} \, dj. \tag{4}$$

Because Korea is a small economy in the model, Korean varieties have measure zero in the U.S. market and do not affect the U.S. industry price index P_{iU}. We assume that $\eta > \sigma$, which implies that the elasticity of substitution among varieties within an industry is larger than the elasticity of demand for the industry composite good.

2.2 Producers

Each industry in China and Korea has a continuum of firms, with the mass of firms normalized to one in each country. Korean imports of Chinese goods are negligible for Chinese industry output Y_i^C and therefore do not affect Chinese marginal cost. We assume that Chinese production exhibits industry-level increasing returns to scale, whereas Korean producers operate under constant returns to scale.[1]

Chinese Producers We assume that production within a Chinese industry exhibits increasing returns to scale as in Kucheryavyy et al. (2023). A Chinese firm j hires labor at wage w^C and its marginal cost is specified as

$$mc_i^C(j) = w^C \left(Y_i^C\right)^{-\alpha}, \tag{5}$$

where Y_i^C is the output of industry i in China and $\alpha > 0$ is the scale elasticity of production. Its marginal cost decreases with the industry-level aggregate output.

The firm exports to the U.S. and Korea and is subject to downward-sloping demand in each destination market. Moreover, it can choose between two different modes of shipping to the U.S. If it chooses to directly ship to the U.S., its goods are subject to ad valorem tariff $\tau \geq 0$, which makes the buyer price and the producer price in the U.S. differ by that margin. On the other hand, it can choose to reroute through a Korean intermediary and bear the rerouting cost, which amounts to a fraction $\varphi_{ij} \in [0, 1)$ of the *revenue*. Each firm is different in terms of its efficiency in rerouting, and it makes

[1] If Korean production also exhibits increasing returns to scale with elasticity $\beta > 0$, Korean marginal cost becomes $mc_i^K = (w^K)^{1-\gamma} P_{iK}^{\gamma} (Y_i^K)^{-\beta}$. The response of Korean output to a change in the U.S. tariff is

$$\frac{d \log Y_i^K}{d\tau} = \frac{\eta - \sigma - \gamma\alpha\sigma\eta}{1 - \eta\beta} \frac{d \log P_{iU}}{d\tau},$$

and increasing returns to scale among Korean producers amplify the response of Korean output to U.S. tariffs if Korean scale elasticity is not too high, i.e., $\eta\beta < 1$. We assume $\gamma\alpha\sigma\eta > \eta - \sigma$ when we analyze the effect of the U.S. tariff on Korean output and demand.

a discrete choice between direct shipment to the U.S. and rerouting through Korea. Its profit-maximization problem can be written as choosing the optimal profit π_{ij} such that

$$\pi_{ij} = \max\left\{\pi_{ij}^{D}, \pi_{ij}^{R}\right\}, \tag{6}$$

$$\pi_{ij}^{D} = \max_{p_{iU}^{CD}(j), p_{iK}^{C}(j)} \left(p_{iU}^{CD}(j) - mc_i^{C}(j)\right)\left((1+\tau)p_{iU}^{CD}(j)\right)^{-\eta} P_{iU}^{\eta-\sigma} + \left(p_{iK}^{C}(j) - mc_i^{C}(j)\right)\left(\frac{p_{iK}^{C}(j)}{P_{iK}}\right)^{-\eta}\left(P_{iK}^{-\sigma} + m_{iK}^{C}\right), \tag{7}$$

$$\pi_{ij}^{R} = \max_{p_{iU}^{CR}(j), p_{iK}^{C}(j)} \left((1-\varphi_{ij})p_{iU}^{CR}(j) - mc_i^{C}(j)\right)\left(p_{iU}^{CR}(j)\right)^{-\eta} P_{iU}^{\eta-\sigma} + \left(p_{iK}^{C}(j) - mc_i^{C}(j)\right)\left(\frac{p_{iK}^{C}(j)}{P_{iK}}\right)^{-\eta}\left(P_{iK}^{-\sigma} + m_{iK}^{C}\right), \tag{8}$$

where the firm takes the U.S. industry price index P_{iU}, the Korean industry price index P_{iK}, and Korean producers' aggregate demand for Chinese intermediate inputs m_{iK}^{C} as given. The optimal producer price for direct shipments to the U.S., the price charged in Korea, and the buyer price for goods rerouted to the U.S. satisfy

$$p_{iU}^{CD}(j) = p_{iK}^{C}(j) = (1-\varphi_{ij})p_{iU}^{CR}(j) = \mu mc_i^{C}(j), \tag{9}$$

where $\mu \equiv \frac{\eta}{\eta-1}$ denotes the constant markup over marginal cost. The firm chooses to reroute through a Korean intermediary if $\pi_{ij}^{R} \geq \pi_{ij}^{D}$, which holds if and only if

$$\varphi_{ij} \leq \frac{\tau}{1+\tau}. \tag{10}$$

Each firm differs in terms of the rerouting cost φ_{ij}, which creates a wedge between the price charged to the U.S. buyer under rerouting and the producer price under direct shipment as

$$\frac{p_{iU}^{CR}(j)}{p_{iU}^{CD}(j)} = \frac{1}{1-\varphi_{ij}} \geq 1.$$

We impose the following assumption on the distribution of firms' rerouting cost φ_{ij}.

Assumption 1. *$1/(1-\varphi_{ij})$ follows the Pareto distribution specified as*

$$Pr\left(\frac{1}{1-\varphi_{ij}} \leq z\right) = 1 - z^{-\kappa_i}, \tag{11}$$

for $\forall z \geq 1$.

A larger κ_i lowers the average rerouting cost faced by Chinese producers. Thus, for a given U.S. tariff, a larger fraction of Chinese firms can reroute through Korea at relatively low cost. Motivated by studies on indirect exporting and shipping costs (Ahn et al. (2011); Ganapati et al. (2024)), we map κ_i in the model to Korean industry i's pre-existing export exposure to the U.S. in the data. This capacity captures the trading relationships and distribution networks that firms in industry i have collectively established in the U.S. market. For example, κ_i is higher for industries with a stronger pre-existing export presence in the U.S.

Korean Producers Korean firms combine labor and the Chinese composite good as production inputs.[2] We assume Korean firms have a Cobb-Douglas production technology with a Chinese-input share of $\gamma \in (0,1)$. To be specific, a Korean firm j in industry i buys $x_{iK}^C(\nu;j)$ units of Chinese variety ν within the same industry i at $p_{iK}^C(\nu)$ and aggregates them into a composite input $x_{iK}^C(j)$ with elasticity of substitution η across varieties as Korean consumers do. Formally,

$$x_{iK}^C(j) = \left(\int_{\nu\in\Omega_{iK}} \left(x_{iK}^C(\nu;j)\right)^{\frac{\eta-1}{\eta}} d\nu\right)^{\frac{\eta}{\eta-1}}. \tag{12}$$

The price of the composite good is P_{iK} in Equation (4) under the assumption that Korean consumers and firms share the same elasticity of substitution across the same set of

[2]We use "final good" only as a reduced-form label for the Korean variety sold in the U.S.; the model does not require a separate final-good sector. The Cobb-Douglas production function directly maps labor and the Chinese composite input into Korean output.

Chinese varieties. A firm's cost minimization yields

$$x_{iK}^{C}(\nu; j) = \left(\frac{p_{iK}^{C}(\nu)}{P_{iK}} \right)^{-\eta} x_{iK}^{C}(j). \tag{13}$$

The marginal cost of a Korean firm j can be written as

$$mc_{i}^{K}(j) = (w^{K})^{1-\gamma} (P_{iK})^{\gamma}, \tag{14}$$

where the firm takes the wage w^K and the price of the Chinese good P_{iK} as given. The Korean firm j faces a downward-sloping demand in the U.S. as

$$q_{iU}^{K}(j) = \left(p_{iU}^{K}(j) \right)^{-\eta} P_{iU}^{\eta-\sigma}.$$

The firm takes the U.S. price index P_{iU} as given. Its optimal price $p_{iU}^{K}(j)$ and its demand for the Chinese input $m_{iK}^{C}(j)$ satisfy

$$p_{iU}^{K}(j) = \mu mc_{i}^{K}(j), \tag{15}$$

$$m_{iK}^{C}(j) = \gamma \frac{p_{iU}^{K}(j) q_{iU}^{K}(j)}{\mu P_{iK}}. \tag{16}$$

2.3 Symmetric Equilibrium

We consider a symmetric equilibrium in which firms in each country choose the same optimal prices for direct shipments to the U.S. Because all Chinese firms share the same production technology and $mc_{i}^{C}(j) = mc_{i}^{C}, \forall j$, they choose the same price for direct shipment to the U.S. as $p_{iU}^{CD}(j) = p_{iU}^{CD}, \forall j$. Likewise, all Korean firms choose the same price as $p_{iU}^{K}(j) = p_{iU}^{K}, \forall j$ because $mc_{i}^{K}(j) = mc_{i}^{K}, \forall j$. However, each Chinese firm's optimal price for goods rerouted through Korea is heterogeneous because $p_{iU}^{CR}(j) = p_{iU}^{CD}/(1 - \varphi_{ij})$. The mass of firms that reroute through Korea is characterized in the following lemma.

Lemma 1. *The mass of Chinese firms that reroute through Korea is*

$$n_i = 1 - (1+\tau)^{-\kappa_i}. \tag{17}$$

For $\tau > 0$, the mass of Chinese firms rerouting via Korea, n_i, increases as the tariff τ and κ_i rise:

$$\frac{dn_i}{d\tau} = \frac{\kappa_i}{1+\tau}(1-n_i) > 0, \tag{18}$$

$$\kappa_i \frac{\partial n_i}{\partial \kappa_i} = -(1-n_i)\log(1-n_i) > 0. \tag{19}$$

At $\tau = 0$ and $n_i = 0$, we have

$$\left.\frac{dn_i}{d\tau}\right|_{\tau=0+} = \kappa_i. \tag{20}$$

Proof. See Appendix A.1. □

A higher U.S. tariff reduces U.S. demand for goods shipped directly from China and makes rerouting through Korea more profitable for a larger mass of Chinese firms. It also increases with the export capacity of the industry in Korea as rerouting through Korea becomes less costly on average. The U.S. import price index is

$$\begin{aligned} P_{iU}^{1-\eta} &= (p_{iU}^{CD})^{1-\eta}\left\{\int_0^{\tau/(1+\tau)}(1-\varphi)^{\eta-1}\kappa_i(1-\varphi)^{\kappa_i-1}d\varphi + (1+\tau)^{1-\eta-\kappa_i}\right\} \\ &= (p_{iU}^{CD})^{1-\eta}\frac{\kappa_i + (\eta-1)(1+\tau)^{-(\kappa_i+\eta-1)}}{\kappa_i+\eta-1}, \end{aligned} \tag{21}$$

and P_{iU} does not include Korean producer prices as Korea is a small economy. Consumption of Chinese goods in the U.S. is given by

$$q_{iU}^C(j) = \begin{cases} [(1+\tau)p_{iU}^{CD}]^{-\eta}P_{iU}^{\eta-\sigma}, & \varphi_{ij} \geq \tau/(1+\tau), \\ [p_{iU}^{CD}/(1-\varphi_{ij})]^{-\eta}P_{iU}^{\eta-\sigma}, & \varphi_{ij} < \tau/(1+\tau). \end{cases} \tag{22}$$

Moreover, the quantity of Chinese goods that are rerouted through Korea is given by

$$R_i = \frac{\kappa_i}{\kappa_i + \eta} \left[1 - (1+\tau)^{-(\kappa_i + \eta)} \right] (p_{iU}^{CD})^{-\eta} P_{iU}^{\eta - \sigma}. \tag{23}$$

Recall that Chinese goods are the only varieties imported by Korean consumers. Hence, the Korean import price index is $P_{iK} = p_{iK}^C = p_{iU}^{CD}$. We denote the total quantity of intermediate inputs imported by Korean firms as $m_{iK}^C \equiv \int x_{iK}^C(j)\, dj$. In symmetric equilibrium, all Korean firms purchase the same amount of intermediate goods. Hence, the quantities of Chinese goods consumed by Korean consumers and inputs used by Korean firms satisfy

$$q_{iK}^C = (p_{iK}^C)^{-\sigma}, \tag{24}$$

$$m_{iK}^C = \gamma \frac{p_i^K q_{iU}^K}{\mu p_{iK}^C}. \tag{25}$$

Total industry output in China and Korea satisfies the market clearing conditions given as

$$Y_i^C = P_{iU}^{-\sigma}, \tag{26}$$

$$Y_i^K = q_{iU}^K = \{\mu (w^K)^{1-\gamma} (p_{iK}^C)^{\gamma}\}^{-\eta} P_{iU}^{\eta - \sigma}. \tag{27}$$

Lastly, Korean total imports and exports satisfy

$$M_i = q_{iK}^C + m_{iK}^C + R_i, \tag{28}$$

$$E_i = Y_i^K + R_i. \tag{29}$$

We establish existence and uniqueness of the symmetric industry equilibrium while assuming that the price elasticity of demand for an industry's composite good is smaller than the elasticity of substitution across varieties within an industry and the scale elastic-

ity of Chinese output is not too high, i.e., $1 < \sigma < \eta < 1/\alpha$. We impose this assumption in the discussion henceforth.

Lemma 2. *Suppose* $1 < \sigma < \eta < 1/\alpha$. *Given* $\tau \geq 0$ *and* $\kappa_i > 0$, *a unique symmetric equilibrium exists.*

Proof. See Appendix A.2. □

2.4 U.S. Tariff on Chinese Goods and Korean Exports and Imports

2.4.1 Economy without Rerouting

We first consider how Korean imports and exports would respond to U.S. tariff hikes in the absence of rerouting. Here, we abstract from Chinese firms' rerouting to isolate the role of increasing returns to scale in Chinese production technology. The economy without rerouting corresponds to the limit case in which $\kappa_i \to 0$ for each finite $\tau \geq 0$, implying $n_i = R_i = 0$. Korean imports of Chinese goods, M_i, are used for final consumption, q_{iK}^C, and production, m_{iK}^C, while Korean exports, E_i, consist of goods produced by Korean firms, Y_i^K, i.e.,

$$M_i = q_{iK}^C + m_{iK}^C, \qquad E_i = Y_i^K. \tag{30}$$

Proposition 1 shows how the contraction in Chinese output, raises the price of Chinese goods in Korea.

Proposition 1. *Without rerouting, a higher U.S. tariff reduces Chinese output and raises the U.S. industry price index. As Chinese production exhibits increasing returns to scale* ($\alpha > 0$), *the contraction in Chinese output also raises the price of Chinese goods in Korea. Korean final consumption of Chinese goods therefore declines. For every finite* $\tau \geq 0$ *and* $\alpha > 0$,

$$\frac{d \log Y_i^C}{d\tau} = -\frac{\sigma}{(1-\alpha\sigma)(1+\tau)} < 0, \quad \frac{d \log P_{iU}}{d\tau} = \frac{1}{(1-\alpha\sigma)(1+\tau)} > 0, \tag{31}$$

$$\frac{d \log p_{iK}^C}{d\tau} = \frac{\alpha\sigma}{(1-\alpha\sigma)(1+\tau)} > 0, \quad \frac{d \log q_{iK}^C}{d\tau} = -\frac{\alpha\sigma^2}{(1-\alpha\sigma)(1+\tau)} < 0. \tag{32}$$

If $\alpha = 0$, the Korean price of Chinese goods and Korean final consumption of Chinese goods do not respond to the U.S. tariff on China:

$$\frac{d \log p_{iK}^C}{d\tau} = 0, \qquad \frac{d \log q_{iK}^C}{d\tau} = 0.$$

Proof. See Appendix A.3. □

The U.S. tariff reduces U.S. purchases of Chinese goods and reduces the scale of Chinese production. With increasing returns to scale in Chinese production ($\alpha > 0$), the contraction in production raises Chinese marginal cost, increasing the price of Chinese goods sold in Korea. The higher price therefore reduces Korean consumption of Chinese final goods. When Chinese production exhibits constant returns to scale ($\alpha = 0$), the price of Chinese goods in Korea and Korean imports of final goods are unchanged by the U.S. tariff. Substitution toward Korean goods by U.S. buyers in turn increases Korean production, exports to the U.S., and intermediate imports from China.

For Korean producers, higher Chinese input prices raise marginal costs, while more expensive Chinese goods in the U.S. shift demand toward Korean varieties. Without rerouting, the response of Korean production, and hence sales to the U.S. market to the tariff hike is obtained as

$$\frac{d \log Y_i^K}{d\tau} = \frac{d \log E_i}{d\tau} = \frac{\eta - \sigma - \gamma\alpha\sigma\eta}{(1 - \alpha\sigma)(1 + \tau)}. \tag{33}$$

The term $\eta - \sigma$ captures substitution away from Chinese goods toward Korean goods, net of the effect of the higher U.S. industry price index, while $\gamma\alpha\sigma\eta$ captures the decline in Chinese input use caused by higher Chinese input prices faced by Korean producers. A higher Chinese-input share, γ, strengthens the pass-through of Chinese input prices into Korean marginal costs, while a higher scale elasticity, α, makes Chinese marginal cost more sensitive to Chinese output. We derive a sufficient condition for Korean imports and exports to decline after the tariff hike.

Proposition 2. *Without rerouting, suppose that*

$$\gamma\alpha\sigma\eta > \eta - \sigma. \tag{34}$$

For every $\tau \geq 0$, an increase in the U.S. tariff reduces Korean output, Y_i^K, and exports to the United States, E_i, as well as Korean imports of Chinese intermediate inputs, m_{iK}^C. Together with the decline in final absorption, this implies a decline in Korean total imports from China, M_i:

$$\frac{dE_i}{d\tau} = \frac{dY_i^K}{d\tau} < 0, \qquad \frac{dm_{iK}^C}{d\tau} < 0, \qquad \frac{dM_i}{d\tau} < 0. \tag{35}$$

Proof. See Appendix A.4. □

The decline in import quantities and rise in import prices depicted in Figure 1 are consistent with higher Chinese marginal costs and their pass-through to Korean production costs through input-output linkages. The condition $\gamma\alpha\sigma\eta > \eta - \sigma$ requires this cost channel to be sufficiently strong: a larger Chinese-input share, γ, strengthens the exposure of Korean producers to Chinese input prices, while a larger scale elasticity, α, makes Chinese marginal costs more responsive to the contraction in Chinese output.

We impose condition (34) when we analyze the economy with rerouting in Section 2.4.2, so that the decline in Korean imports from China due to higher Chinese marginal costs can be weighed against the increase in imports induced by rerouting, allowing responses to vary across sectors.

2.4.2 Economy with Rerouting

Incorporating Chinese firms' rerouting through Korea generates heterogeneous responses across sectors in Korean imports from China and exports to the U.S. when the U.S. tariff on China rises. Rerouting introduces an additional channel through which the U.S. tariff hike affects Korea, counteracting the decrease in Korean imports from China

explored in Section 2.4.1. The tariff increase encourages Chinese goods to be rerouted through Korea, thereby increasing Korean imports from China. Which effect dominates depends on the industry's average rerouting cost, governed by $\kappa_i > 0$.

The tariff has opposing effects on the extensive and intensive margins of Chinese rerouting. On the extensive margin, a higher U.S. tariff increases the mass of Chinese firms rerouting through Korea. On the intensive margin, however, firms that were already rerouting reduce the quantities they ship, reflecting higher marginal costs due to lower U.S. import demand following the tariff increase:

$$\frac{d \log q_{iU}^{C}(j)}{d\tau} = (\eta - \sigma - \alpha\sigma\eta) \frac{d \log P_{iU}}{d\tau} < 0. \tag{36}$$

Total rerouted volume R_i can therefore increase or decrease, depending on whether the additional shipments from newly rerouting firms outweigh the decline in shipments by existing rerouters. Korean total imports from China increase only if the increase in rerouted trade is large enough to offset the declines in Korean final demand for Chinese goods and imports of Chinese intermediate inputs. Proposition 3 shows how Korean industry i's average rerouting cost, which may vary with its pre-existing export capacity to the United States, determines this balance.

Proposition 3 (Korean imports from China)**.** *Let $M_i = q_{iK}^{C} + m_{iK}^{C} + R_i$ denote Korean imports from China, where q_{iK}^{C} is Korean final absorption, m_{iK}^{C} is Korean intermediate absorption, and R_i is the quantity of Chinese goods rerouted through Korea. Let κ_i measure Korean industry i's export capacity to the U.S. For every $\tau \geq 0$, there is a unique $\kappa_i^{M}(\tau) \in (0, \infty)$ such that*

$$\frac{dM_i}{d\tau} < 0 \text{ if } \kappa_i < \kappa_i^{M}(\tau), \qquad \frac{dM_i}{d\tau} = 0 \text{ if } \kappa_i = \kappa_i^{M}(\tau), \qquad \frac{dM_i}{d\tau} > 0 \text{ if } \kappa_i > \kappa_i^{M}(\tau). \tag{37}$$

At low export capacity, falling Korean absorption and lower sales by incumbent rerouters dominate. At high export capacity, the entry of additional Chinese firms into rerouting dominates those contractions.

Proof. See Appendix A.5. □

The proposition shows that the response of Korean imports from China depends on the extent of additional rerouting induced by a higher tariff. When κ_i is low, i.e., when the rerouting cost is high on average, relatively few Chinese firms begin rerouting through Korea. Therefore, the declines in Korean demand for Chinese goods and in shipments by existing rerouters dominate. When κ_i is high, the additional shipments associated with newly rerouting firms are large enough to offset these declines. Thus, an increase in Korean imports from China following a U.S. tariff increase need not reflect stronger Korean demand for Chinese goods, but may instead be driven by an increase in Chinese goods rerouted through Korea.

We next consider Korean exports to the United States. An increase in the U.S. tariff reduces Korean-produced exports by raising the cost of Chinese intermediate inputs, while at the same time increasing the volume of Chinese goods rerouted through Korea to the U.S. Proposition 4 shows how the Korean industry's export capacity determines the response of total Korean exports to the United States.

Proposition 4 (Korean exports to the U.S.)**.** *Let $E_i = Y_i^K + R_i$ denote Korean total exports to the U.S., where Y_i^K is Korean-produced output and R_i is the quantity of Chinese goods rerouted through Korea. For every $\tau \geq 0$, there is a unique $\kappa_i^E(\tau) \in (0, \infty)$ such that*

$$\frac{dE_i}{d\tau} < 0 \text{ if } \kappa_i < \kappa_i^E(\tau), \qquad \frac{dE_i}{d\tau} = 0 \text{ if } \kappa_i = \kappa_i^E(\tau), \qquad \frac{dE_i}{d\tau} > 0 \text{ if } \kappa_i > \kappa_i^E(\tau). \tag{38}$$

At low export capacity, the contraction in Korean-produced exports dominates. At high export capacity, additional rerouted Chinese goods more than offset the contraction in Korean production.

Proof. See Appendix A.6. □

Similarly, Korean total exports to the United States can increase when the additional Chinese goods rerouted through Korea more than offset the decline in Korean-produced

exports. Proposition 4 shows that this occurs when κ_i is sufficiently high. Thus, an increase in Korean total exports following a U.S. tariff increase need not reflect an expansion in production by Korean firms.

We conclude this section with four predictions that we test using Korean customs data. First, Propositions 1 and 2 imply that, absent rerouting, a higher U.S. tariff reduces Korean final and intermediate imports.

Prediction 1. *U.S. tariffs on China reduce Korean imports from China, absent rerouting, with larger declines in industries where Chinese production exhibits increasing returns to scale and in products with greater exposure to Chinese imported inputs.*

Second, Proposition 1 sheds light on the mechanism behind the effect of the U.S. tariff on Korean imports from China. The U.S. tariff raises the marginal cost of Chinese producers and the import price of Chinese goods in Korea.

Prediction 2. *U.S. tariffs on China raise the import prices of Chinese goods in Korea.*

Third, Proposition 3 shows that rerouting of Chinese goods through Korea may offset the effect of the U.S. tariffs on Korean imports depending on the average rerouting cost faced by Chinese producers.

Prediction 3. *The decline in Korean imports from China is weaker, or may reverse, in industries with low costs of rerouting through Korea.*

Fourth, Proposition 4 gives the corresponding prediction for Korean exports to the United States. At low rerouting costs, additional rerouted Chinese goods can more than offset the decline in Korean-produced exports.

Prediction 4. *U.S. tariffs on China reduce Korean exports to the United States absent rerouting, but the effect may reverse in industries with low costs of rerouting through Korea.*

In the model, a larger κ_i lowers the average rerouting cost faced by Chinese producers. Thus, for a given U.S. tariff, a larger fraction of Chinese firms can reroute through

Korea at relatively low cost. Following the discussion in Section 2.2, we map κ_i in the model to Korean industry i's pre-existing U.S.-export exposure in the data. We verify these predictions using Korean customs data in the following section.

3 Empirical Analysis

3.1 Data

We employ transaction-level Korean customs data. The dataset encompasses the universe of Korean import and export transactions reported to the Korea Customs Service from January 2014 through December 2019. Each transaction is recorded at the HS10 product – source/origin country – monthly date level, where products are classified at the 10-digit Harmonized System of Korea (HSK) — the most granular product classification available for Korean trade.

In the empirical analysis, we aggregate the transaction-level records to an annual frequency. Each observation in the panel is at the HS10 product – origin country – year level. Each observation includes the import value and weight, which determine the unit value per kilogram. Following the convention in the literature, we use the weight in kilograms as our proxy for trade quantity and the unit value as our proxy for the import price. Unless otherwise noted, regressions are weighted by the corresponding trade quantity, using import weight for import regressions and export weight for export regressions.

To measure the bilateral tariff escalation between the U.S. and China in 2018 and 2019, we use product-level tariff changes constructed by Fajgelbaum et al. (2020) and Bown (2021a). For each HS6 product i, we compute $d\tau_i^{US-CN}$ as the cumulative change in the U.S. statutory import tariff on Chinese products between 2017 and 2019, and $d\tau_i^{CN-US}$ as the analogous cumulative change in China's statutory import tariff on U.S. products over the same window. We map these HS6-level tariff changes to Korea's HS10 product codes, so that every HS10 product in the Korean customs panel is associated with bilateral

tariff changes capturing, respectively, the intensity of the U.S. tariff hike on China and the intensity of China's retaliation against the U.S.

On average, Korean firms import from approximately 240 origin countries and source about 11,000 distinct HS10 products per year. China plays a prominent role in Korean imports. Korean firms import about 9,200 distinct HS10 products from China each year, and as of 2017, China accounted for 20.5% of Korea's total import value. Korean imports from China are concentrated in a few HS2 chapters: the top three chapters — electrical machinery (HS85, 33.2%), nuclear reactors and machinery (HS84, 13.5%), and iron and steel (HS72, 7.2%) — account for 53.9% of total imports from China in 2017, and the top ten chapters account for 74.7%. Imports of intermediate goods account for a sizable portion of imports from China in 2017, amounting to 62.8% of the import value of the top ten chapters.

TABLE 1: Top 10 HS2 Chapters of Korean Imports from China, 2017

Rank	HS2	Description	Share	Intermediate Goods Share
1	85	Electrical machinery	33.2%	67.8%
2	84	Nuclear reactors, machinery	13.5%	31.4%
3	72	Iron and steel	7.2%	99.6%
4	90	Optical, photographic instruments	4.3%	19.8%
5	73	Articles of iron or steel	3.4%	88.7%
6	29	Organic chemicals	3.2%	100.0%
7	28	Inorganic chemicals	3.0%	100.0%
8	39	Plastics	2.7%	83.1%
9	62	Articles of apparel and clothing accessories	2.1%	0.0%
10	94	Furniture	2.0%	34.8%
		Top 10 share	74.7%	62.8%

Notes: HS2 chapter shares of total Korean imports from China in 2017. Total Korean imports from China in 2017 are $97.9 billion. Source: Korean customs data from KTSPI. The intermediate goods share is the value share of intermediate goods, classified by BEC Revision 4, in each HS2 chapter's imports from China. Description columns abbreviate the official HS2 chapter names.

About 92.7% of HS10 products imported from China were subject to U.S. tariff increases between January 2018 and December 2019, and about 87.3% of HS10 products imported from the U.S. were subject to China's retaliatory tariffs over the same window.

3.2 Empirical Analysis: Evidence of Decoupling from China

We first document how Korean imports from China responded to the U.S. tariff shock at the product level. To analyze how the U.S.–China tariff shocks transmitted to Korean imports at the product level, we estimate the following difference-in-differences (DID) specification at the HS10 product × origin country × year level:

$$\begin{aligned} y_{i,c,t} \;=\; & \alpha + \beta_P \, Post_t + \beta_1 \, d\tau_i^{US-CN} + \beta_2 \, d\tau_i^{CN-US} \\ & + \gamma_{US} \, D_c^{US} \, Post_t + \gamma_{CN} \, D_c^{CN} \, Post_t \\ & + \delta_{USCN} \, D_c^{CN} \, d\tau_i^{US-CN} + \delta_{CNUS} \, D_c^{US} \, d\tau_i^{CN-US} \\ & + \theta_{USCN} \, Post_t \, d\tau_i^{US-CN} + \theta_{CNUS} \, Post_t \, d\tau_i^{CN-US} \\ & + \phi_{USCN} \, D_c^{CN} \, Post_t \, d\tau_i^{US-CN} + \phi_{CNUS} \, D_c^{US} \, Post_t \, d\tau_i^{CN-US} \\ & + \alpha_{s(i),c} + \alpha_{s(i),t} + \varepsilon_{i,c,t}, \end{aligned} \tag{39}$$

where the outcome $y_{i,c,t}$ is the log import quantity of HS10 product i imported from origin country c in year t. $s(i)$ denotes the HS4 digit category of the product i. $d\tau_i^{US-CN}$ denotes the cumulative change in the U.S. import tariff on Chinese HS6 product i between 2017 and 2019. $d\tau_i^{CN-US}$ is the analogous change in China's import tariff on U.S. goods of product i over the same window, capturing the intensity of China's retaliation against U.S. exports. $Post_t$ is an indicator equal to one for years from 2018 onward, and D^{US} and D^{CN} are origin-country dummies for the U.S. and China, respectively. The fixed effects $\alpha_{s(i),c}$ and $\alpha_{s(i),t}$ absorb HS4 product × origin country characteristics, as well as product-specific year effects. Our key coefficients of interest are ϕ_{USCN} and ϕ_{CNUS}. The coefficient on $d\tau_i^{US-CN} \times Post_t \times D^{CN}$ measures how Korean imports from China responded to U.S. tariffs; the coefficient on $d\tau_i^{CN-US} \times Post_t \times D^{US}$ measures how Korean imports from the U.S. responded to China's retaliatory tariffs at the product level. Standard errors are

clustered at the HS4 product level.[3]

Table 2 reports estimates of Equation (39), with log import weight as the outcome variable. The coefficient on $D_c^{CN}\, Post_t\, d\tau_i^{US-CN}$ is $\hat{\phi}_{USCN} = -2.37$, statistically significant at the 1% level. That is, a 10 percentage-point increase in the U.S. tariff on Chinese product i leads to roughly a 24% decline in Korean import quantity of that product from China after 2017.[4] The negative import response is consistent with Prediction 1, which summarizes the model's implication that U.S. tariffs on China reduce Korean imports from China absent rerouting.

The coefficient on $D_c^{US}\, Post_t\, d\tau_i^{CN-US}$, capturing the response of Korean imports from the U.S., is *not statistically different from zero*. The estimate on $D_c^{US}\, Post_t$ in Column 1 of Table 2 is positive, +1.320, statistically significant at the 10% level: on average, Korean imports from the U.S. rose after 2017. Thus, while Korean imports from the U.S. increased on average, this increase is not systematically concentrated in products facing larger Chinese retaliatory tariff hikes. The estimates are quantitatively consistent with the aggregate patterns documented in the Introduction: Korean import quantity from China fell after 2017, shown in Figure 1.

The empirical pattern documented in our paper stands in contrast to the rerouting of Chinese products through bystander countries documented for Mexico and Vietnam (Utar et al., 2025b; Iyoha et al., 2024), and points to important cross-country heterogeneity in how bystander countries adjusted their imports from China during the trade war. Testing our model predictions 1 and 2, we investigate the mechanism behind the decline in Korean imports from China in Section 3.3.

We have conducted some additional exercises to show that the decline in imports from China is robust across different specifications. First, the tariff measure is non-zero for

[3]The estimation sample restricts to observations with the annual log change in weight between -log 2 and log 2. All regressions are weighted by import weight, so that the estimated coefficient reflects a quantity-weighted product-level response rather than an unweighted average.

[4]About 74% of products affected by the U.S. tariff face a tariff of 25% or higher, with a median of 25%. For these products, the implied total post-2017 decline in Korean imports from China is roughly 13 to 25%.

TABLE 2: Baseline DID

	Dependent variable: Import Quantity	
	(1)	(2)
CN × Post × $\tau^{\text{US-CN}}$	-2.373***	
	(0.893)	
US × Post × $\tau^{\text{CN-US}}$	-4.119	
	(2.829)	
US × Post	1.320*	0.373*
	(0.748)	(0.224)
CN × Post	0.465**	-0.183*
	(0.233)	(0.097)
$\tau^{\text{US-CN}}$	0.401	
	(1.232)	
$\tau^{\text{CN-US}}$	-0.629	
	(3.103)	
CN × $\tau^{\text{US-CN}}$	0.517	
	(2.386)	
US × $\tau^{\text{CN-US}}$	4.752	
	(4.606)	
Fixed Effects	HS4 x Year, HS4 x Country	
Observations	658,192	658,562

Notes: Each observation is at the HS10 product × origin country × year level. All regressions are weighted by import quantities, proxied by their weights in kilograms. The estimation sample restricts to observations with the annual log change in weight between −2 and 2. We include HS4 × year and HS4 × country fixed effects. Standard errors clustered at HS4 in parentheses. * p<0.1, ** p<0.05, *** p<0.01.

most HS10 products, so we instead look at how Korean imports from the U.S. and from China have been impacted on average across all HS10 products, abstracting away from the cross-sectional variation in $d\tau$ across HS10 products. We estimate the below specification, leveraging the variation across source countries only:

$$y_{i,c,t} = \alpha + \beta_P \, Post_t + \gamma_{US} \, D_c^{US} \, Post_t + \gamma_{CN} \, D_c^{CN} \, Post_t + \alpha_{s(i),c} + \alpha_{s(i),t} + \varepsilon_{i,c,t}. \tag{40}$$

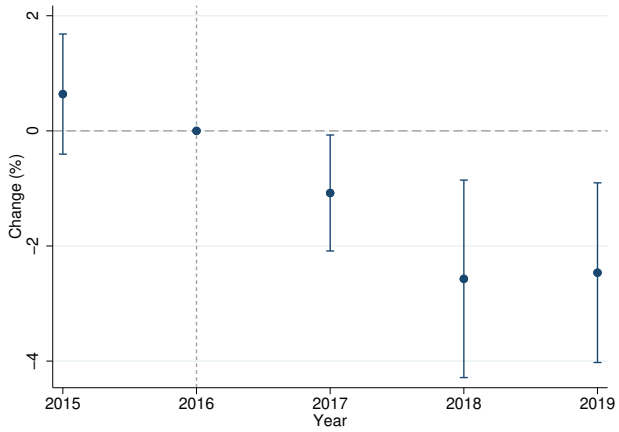
Column 2 of Table 2 reports estimates of Equation (40). On average, Korean imports from the U.S. rose by 37% after 2017, and Korean imports from China fell by 18%. Both estimates are statistically different from zero at the 10% level. The results are qualitatively similar to those in Column 1, reaffirming our main findings.

Second, we estimate a dynamic event-study version of Equation (39), replacing the $Post_t$ indicator with year dummies, taking 2016 as a reference year. Letting $\mathcal{K} = \{2015, 2017, 2018, 2019\}$ and letting D_t^k be a dummy equal to one if $t = k$, the cross-only dynamic spec-

ification is

$$
\begin{aligned}
y_{i,c,t} \;=\; & \alpha + \sum_{k\in\mathcal{K}} \beta^k \, D_t^k + \beta_1 \, d\tau_i^{US-CN} + \beta_2 \, d\tau_i^{CN-US} \\
& + \sum_{k\in\mathcal{K}} \left(\gamma^{US,k} D_c^{US} \, D_t^k + \gamma^{CN,k} D_c^{CN} \, D_t^k \right) \\
& + \delta_{USCN} \, D_c^{CN} \, d\tau_i^{US-CN} + \delta_{CNUS} \, D_c^{US} \, d\tau_i^{CN-US} \\
& + \sum_{k\in\mathcal{K}} \left(\theta^k_{USCN} \, D_t^k \, d\tau_i^{US-CN} + \theta^k_{CNUS} \, D_t^k \, d\tau_i^{CN-US} \right) \\
& + \sum_{k\in\mathcal{K}} \left(\phi^k_{USCN} \, D_c^{CN} \, D_t^k \, d\tau_i^{US-CN} + \phi^k_{CNUS} \, D_c^{US} \, D_t^k \, d\tau_i^{CN-US} \right) \\
& + \alpha_{s(i),c} + \alpha_{s(i),t} + \varepsilon_{i,c,t}. \qquad (41)
\end{aligned}
$$

FIGURE 2. Dynamic Effects of U.S. Tariffs on Korean Imports from China



Notes: The figure plots the year-by-year coefficients $\{\hat{\phi}^k_{USCN}\}_{k\in\mathcal{K}}$ estimated from Equation (41), capturing the year-by-year response of Korean imports from China to the U.S. tariff on China, relative to the 2016 base year. The bars are 90% confidence intervals using standard errors clustered by HS4 digit levels. * p<0.1, ** p<0.05, *** p<0.01.

Figure 2 shows the regression results of Equation (41) with log import weight as the dependent variable. The coefficients of interest are $\{\hat{\phi}^k_{USCN}\}_{k\in\mathcal{K}}$, capturing the year-by-year response of Korean imports from China to the U.S. tariff on China, relative to the 2016 base year. The coefficient estimates $\hat{\phi}^k_{USCN}$ are *not statistically different from zero* before 2018. That is, we could not find any evidence of systematic differences in the quantity

adjustment of Korean imports from China across products with different exposures to the U.S. tariff on China, prior to the trade war. The negative effect of the U.S. tariff on Korean imports from China becomes evident after the trade war: the estimate $\hat{\phi}^k_{USCN}$ becomes negative and statistically different from zero, starting in 2018. The estimates in 2018 and 2019 are quantitatively consistent with the static estimate of $\hat{\phi}_{USCN} = -2.37$ reported in Column 1 of Table 2.

Third, Table 8 in the Appendix replicates Equation (39) with nine different fixed-effects included in the estimation. The estimate on ϕ_{USCN} remains negative and statistically significant at the 1% level across all nine specifications, ranging from -1.34 to -2.64. Across these checks, the decline in Korean imports from China after 2017 is robust to alternative specifications and inclusion of different sets of fixed effects.

3.3 Inspecting the Mechanism

3.3.1 The Role of Increasing Returns to Scale

We next examine whether the decline in Korean imports from China is concentrated in industries where the model predicts a stronger response. We divide HS10 products into those produced in industries exhibiting increasing returns to scale (IRS) and those produced in constant-returns-to-scale (CRS) industries, and examine whether the effects of U.S. tariffs differ across the two groups. The IRS classification follows Antweiler and Trefler (2002), who classify ISIC Revision 2 industries according to their returns-to-scale characteristics. We map their industry-level classification to HS products using the concordance table provided by the World Bank.[5]

Let $IRS_i \in \{0, 1\}$ indicate whether product i is produced in an IRS industry, and let $CRS_i = 1 - IRS_i$ denotes the corresponding indicator for a CRS industry. We allow the

[5] Antweiler and Trefler (2002) distinguish among CRS industries, robustly classified IRS industries, and IRS industries whose classification is not robust across specifications. In our baseline analysis, we classify all industries not identified as CRS—including the non-robust IRS group—as IRS. The results remain robust, however, to a more restrictive definition that excludes the non-robust IRS industries.

effect of the U.S. tariff to differ between the two groups by estimating

$$\begin{aligned} y_{i,c,t} =& \Theta^{US}_{i,c,t} + \lambda_0\, IRS_i + \lambda_1\, D^{CN}_c IRS_i + \lambda_2\, Post_t IRS_i + \Gamma^{IRS}_{i,c,t}\, IRS_i + \Gamma^{CRS}_{i,c,t}\, CRS_i \\ &+ \alpha_{s(i),c} + \alpha_{s(i),t} + \varepsilon_{i,c,t}, \end{aligned} \tag{42}$$

where $\Theta^{US}_{i,c,t}$ collects the terms of Equation (39) that involve China's retaliatory tariff, $d\tau_i^{CN-US}$, or the U.S.-origin dummy, D^{US}_c:

$$\begin{aligned} \Theta^{US}_{i,c,t} = \alpha &+ \beta_2\, d\tau_i^{CN-US} + \gamma_{US}\, D^{US}_c\, Post_t \\ &+ \delta_{CNUS}\, D^{US}_c\, d\tau_i^{CN-US} + \theta_{CNUS}\, Post_t\, d\tau_i^{CN-US} + \phi_{CNUS}\, D^{US}_c\, Post_t\, d\tau_i^{CN-US}, \end{aligned}$$

and the triple interaction $D^{CN}_c\, Post_t\, d\tau_i^{US-CN}$, together with all of its lower-order terms, enters separately for products exhibiting IRS and CRS:

$$\begin{aligned} \Gamma^{g}_{i,c,t} = \lambda^g_3\, d\tau_i^{US-CN} &+ \lambda^g_4\, D^{CN}_c\, Post_t + \lambda^g_5\, D^{CN}_c\, d\tau_i^{US-CN} \\ &+ \theta^g_{USCN}\, Post_t\, d\tau_i^{US-CN} + \phi^g_{USCN}\, D^{CN}_c\, Post_t\, d\tau_i^{US-CN}, \qquad g \in \{IRS, CRS\}. \end{aligned}$$

Table 3 examines whether the effect of U.S. tariffs on Korean imports from China varies with the returns-to-scale characteristics of the affected products. Column (1) reports estimates from the full sample, allowing the tariff effect to differ between products in constant-returns-to-scale (CRS) and increasing-returns-to-scale (IRS) industries. The results show a sharp contrast between the two product groups. For CRS products, the estimated effect is positive but small and statistically insignificant. By contrast, the effect for IRS products is negative, large, and statistically significant. Specifically, among IRS products, a 10-percentage-point increase in U.S. tariffs on Chinese goods is associated with a 28 percent decline in Korea's import quantity from China in the post-2017 period. Columns (2) and (3) report estimates obtained by dividing the sample into CRS and IRS products, respectively. The estimated tariff effects in these subsamples are nearly identi-

TABLE 3: Heterogeneity by Returns to Scale

	Dependent variable: Import Quantity		
	(1) Full Sample	(2) CRS Sample	(3) IRS Sample
CN × Post × $\tau^{\text{US-CN}} \times IRS_i$	-2.7971**		-2.8264**
	(1.1115)		(1.1174)
Post × $\tau^{\text{US-CN}} \times IRS_i$	0.5753		0.6271
	(0.9501)		(0.9543)
CN × Post × $\tau^{\text{US-CN}} \times CRS_i$	0.7406	0.3389	
	(1.3242)	(1.4024)	
Post × $\tau^{\text{US-CN}} \times CRS_i$	0.7905	0.5044	
	(0.5507)	(0.4047)	
US × Post × $\tau^{\text{CN-US}}$	-4.5001	-0.2429	-5.3481
	(2.9670)	(0.5965)	(3.2841)
US × Post	1.4445*	0.3306	1.6391*
	(0.7890)	(0.2259)	(0.8470)
Fixed Effects	HS4 x Year, HS4 x Country		
Observations	599,298	161,473	437,743

Notes: Each observation is at the HS10 product × origin country × year level. We include HS4 × year and HS4 × country fixed effects. Standard errors clustered at HS4 in parentheses. IRS_i is equal to one if HS10 product i is produced in an industry classified as exhibiting increasing returns to scale, and $CRS_i = 1 - IRS_i$, following the classification of Antweiler and Trefler (2002). In column (1), the effect for each group is given directly by the coefficient on CN × Post × $\tau^{\text{US-CN}} \times IRS_i$ or CN × Post × $\tau^{\text{US-CN}} \times CRS_i$. In columns (2) and (3), the corresponding effects are estimated separately on the CRS and IRS subsamples. Other regressors are suppressed for brevity. * p<0.1, ** p<0.05, *** p<0.01.

cal in magnitude to the corresponding effects from the full-sample specification in column (1), reinforcing the finding that the decline in Korean imports from China is concentrated among IRS products.

These findings suggest that the decline in Korean imports from China was concentrated in tariff-exposed IRS industries. U.S. tariffs reduced Chinese production in these industries, and the resulting loss of scale economies raised unit costs, reducing Korea's imports from China. This concentration of import declines in IRS industries echoes Prediction 1.

3.3.2 Price Response

We next examine whether the decline in Korean import quantities from China is accompanied by higher prices for Chinese goods in Korea. Table 4 reports the estimates of Equations (39) and (40) using log unit value, our proxy for import price, as the outcome. The coefficient on $D_c^{CN}\, Post_t\, d\tau_i^{US-CN}$ is 0.245, statistically significant at the 10%

level. That is, a 10 percentage-point increase in the U.S. tariff on Chinese product i leads to roughly a 2.5% increase in the Korean import price of that product from China after 2017. The tariff-exposure coefficient for U.S. imports is *not statistically different from zero*. This price response is in line with Prediction 2, which links the decline in Korean import quantities from China to higher prices for Chinese goods in Korea.

The combination of falling quantity ($\hat{\phi}_{USCN} < 0$) and rising price ($\hat{\phi}^{\text{price}}_{USCN} > 0$) for Korean imports from China is consistent with a negative supply shock propagating from China. As the U.S. tariff reduced U.S. demand for Chinese exporters' products, their marginal costs may have risen if production faces increasing returns to scale. This higher marginal cost may have put upward pressure on the prices Korea paid for the Chinese products that did ship, reducing the quantity Korea imports from China. A rerouting story, in which Chinese goods are diverted from the U.S. to Korea, would have predicted the opposite signs: rising quantities and falling prices. The Korean evidence for imports from China therefore departs from the rerouting narrative documented for Mexico and Vietnam (Utar et al., 2025b; Iyoha et al., 2024). For Korean imports from the U.S., the rise in volume with little price movement is consistent with U.S. exporters redirecting products toward Korea at roughly unchanged prices, as documented by Hardy and Saffie (2024).

3.3.3 Input share

We next examine how higher Chinese input prices affect Korean trade through input–output linkages. In the model, U.S. tariffs reduce Chinese production scale, raise Chinese marginal costs, and increase the price of Chinese goods supplied to Korea. For Korean producers, higher Chinese input prices raise marginal costs. When the Chinese-input share is sufficiently large, this pass-through to Korean marginal costs is strong enough to reduce Korean output and, in turn, demand for Chinese intermediate inputs.

We first investigate whether the effect of U.S. tariffs on Korean imports from China

TABLE 4: Effects on Korean Import Prices

	Dependent variable: Price	
	(1)	(2)
CN × Post × $\tau^{\text{US-CN}}$	0.245*	
	(0.125)	
US × Post × $\tau^{\text{CN-US}}$	0.039	
	(0.139)	
US × Post	-0.048*	-0.038
	(0.027)	(0.025)
CN × Post	-0.004	0.060**
	(0.051)	(0.026)
$\tau^{\text{US-CN}}$	-0.371	
	(1.244)	
$\tau^{\text{CN-US}}$	0.402	
	(0.475)	
CN × $\tau^{\text{US-CN}}$	-1.260	
	(1.191)	
US × $\tau^{\text{CN-US}}$	-0.778	
	(1.407)	
Fixed Effects	HS4 x Year, HS4 x Country	
Observations	658,192	658,562

Notes: Each observation is at the HS10 product × origin country × year level. We include HS4 × year and HS4 × country fixed effects. Standard errors clustered at HS4 in parentheses. * p<0.1, ** p<0.05, *** p<0.01.

is more pronounced for products with greater exposure to Chinese imported inputs. For each HS6 category, we construct a measure of exposure to Chinese imported inputs from the input-output table. The measure allocates imported intermediate input use to HS6 categories and weights it by China's share of Korean imports within the corresponding HS6 category. We compute this measure at the HS6 level, the most disaggregated level at which HS products can be matched to input–output categories. An HS10 product is classified as having high exposure to Chinese imported inputs if the share for its HS6 category exceeds the 75th percentile across HS6 categories, and as having low exposure to Chinese imported inputs otherwise.

Let $S_i^{CN,H} \in \{0,1\}$ indicate high exposure to Chinese imported inputs and $S_i^{CN,L} \equiv 1 - S_i^{CN,H}$ low exposure. We allow the effect of the U.S. tariff to differ between the two groups by estimating

$$y_{i,c,t} = \Theta_{i,c,t}^{US} + \lambda_0 S_i^{CN,H} + \lambda_1 D_c^{CN} S_i^{CN,H} + \lambda_2 Post_t S_i^{CN,H} + \Gamma_{i,c,t}^{H} S_i^{CN,H} + \Gamma_{i,c,t}^{L} S_i^{CN,L}$$

$$+ \alpha_{s(i),c} + \alpha_{s(i),t} + \varepsilon_{i,c,t}, \tag{43}$$

where $\Theta^{US}_{i,c,t}$ collects the terms of Equation (39) that involve China's retaliatory tariff, $d\tau_i^{CN-US}$, or the U.S.-origin dummy, D_c^{US}:

$$\begin{aligned}\Theta^{US}_{i,c,t} = {} & \alpha + \beta_2\, d\tau_i^{CN-US} + \gamma_{US}\, D_c^{US}\, Post_t \\ & + \delta_{CNUS}\, D_c^{US}\, d\tau_i^{CN-US} + \theta_{CNUS}\, Post_t\, d\tau_i^{CN-US} + \phi_{CNUS}\, D_c^{US}\, Post_t\, d\tau_i^{CN-US},\end{aligned}$$

and the triple interaction $D_c^{CN}\, Post_t\, d\tau_i^{US-CN}$, together with all of its lower-order terms, enters separately for products with high ($g = H$) and low ($g = L$) exposure to Chinese imported inputs:[6]

$$\begin{aligned}\Gamma^{g}_{i,c,t} = {} & \lambda_3^g\, d\tau_i^{US-CN} + \lambda_4^g\, D_c^{CN}\, Post_t + \lambda_5^g\, D_c^{CN}\, d\tau_i^{US-CN} \\ & + \theta^g_{USCN}\, Post_t\, d\tau_i^{US-CN} + \phi^g_{USCN}\, D_c^{CN}\, Post_t\, d\tau_i^{US-CN}, \qquad g \in \{H, L\}.\end{aligned}$$

The effect of the U.S. tariff on Korean imports from China is thus ϕ^H_{USCN} for products with high exposure and ϕ^L_{USCN} for products with low exposure.

Table 5 reports the regression results for the two groups. We find that the decline in Korean imports from China following an increase in U.S. tariffs is statistically significant and concentrated among products with high exposure to Chinese imported inputs. Specifically, a 10 percentage point increase in U.S. tariffs on Chinese goods is associated with a decline of around 25 percent in Korean import quantity from China among products with high exposure to Chinese imported inputs. For products with low exposure to Chinese imported inputs, the estimated effect is not statistically significant. This concentration of import declines among products with greater exposure to Chinese imported inputs echoes the Chinese-input dimension of Prediction 1.

[6] $Post_t$ and D_c^{US} do not enter $\Theta^{US}_{i,c,t}$ separately because they are absorbed by the HS4 × year and HS4 × country fixed effects, $\alpha_{s(i),t}$ and $\alpha_{s(i),c}$.

TABLE 5: Heterogeneity by China import share: Import Quantity Responses

	Dependent variable: Import Quantity
	(1)
CN × Post × $\tau^{\text{US-CN}} \times S_i^{\text{CN,H}}$	-2.5735***
	(0.8908)
Post × $\tau^{\text{US-CN}} \times S_i^{\text{CN,H}}$	1.2663
	(1.2364)
CN × Post × $\tau^{\text{US-CN}} \times S_i^{\text{CN,L}}$	-0.6116
	(1.5553)
Post × $\tau^{\text{US-CN}} \times S_i^{\text{CN,L}}$	-0.6671
	(0.8765)
US × Post × $\tau^{\text{CN-US}}$	-4.0332
	(2.8390)
US × Post	1.2989*
	(0.7520)
Fixed Effects	HS4 x Year, HS4 x Country
Observations	658,192

Notes: Each observation is at the HS10 product × origin country × year level. We include HS4 × year and HS4 × country fixed effects. Standard errors clustered at HS4 in parentheses. $S_i^{\text{CN,H}}$ is equal to one if imported HS10 product i belongs to an HS6 category whose exposure to Chinese imported inputs exceeds the 75th percentile across HS6 categories, and $S_i^{\text{CN,L}} = 1 - S_i^{\text{CN,H}}$. Other regressors are suppressed for brevity. * p<0.1, ** p<0.05, *** p<0.01.

Overall, the import responses support the cost-pass-through channel emphasized above. U.S. tariffs raise Chinese marginal costs and the prices of Chinese goods supplied to Korea. Higher Chinese input prices then raise Korean producers' marginal costs and reduce Korean output, which in turn lowers Korea's demand for Chinese intermediate inputs, particularly for products with greater exposure to Chinese imported inputs.

3.4 Heterogeneity by Pre-shock U.S.-Export Exposure: Evidence for Rerouting of Chinese Imports to the U.S.

The aggregate decline in Korean imports from China need not imply a uniform decline across products. The model predicts that rerouting through Korea can offset or reverse this decline in industries where Chinese producers face lower rerouting costs. To examine how the response of Korean imports from China varies with Korean exporters' pre-shock exposure to the U.S. market, we add a heterogeneity layer to Equation (39). The U.S. tariff on China is interacted with a pre-shock indicator $S_i^{US,H} \in \{0,1\}$, equal to one if

HS10 product i belongs to an HS2 industry in the top decile of the share of its total exports shipped to the U.S. in 2017 before the trade war.[7] We write $S_i^{US,L} \equiv 1 - S_i^{US,H}$ for the indicator of low U.S. export exposure. Motivated by studies on indirect exporting and shipping costs (Ahn et al. (2011); Ganapati et al. (2024)), we use the pre-shock share of Korean industry exports shipped to the U.S. to proxy for the model's rerouting-cost parameter. A larger U.S.-export share captures stronger trading relationships and distribution networks in the U.S. market, which may lower the cost of routing Chinese goods through Korea. Specifically, we estimate:

$$y_{i,c,t} \;=\; \Theta_{i,c,t}^{US} + \Gamma_{i,c,t}^{H}\, S_i^{US,H} + \Gamma_{i,c,t}^{L}\, S_i^{US,L} + \alpha_{s(i),c} + \alpha_{s(i),t} + \varepsilon_{i,c,t}, \tag{44}$$

where $\Theta_{i,c,t}^{US}$ collects the terms of Equation (39) that involve China's retaliatory tariff, $d\tau_i^{CN-US}$, or the U.S.-origin dummy, D_c^{US}:

$$\begin{aligned}\Theta_{i,c,t}^{US} = \alpha &+ \beta_2\, d\tau_i^{CN-US} + \gamma_{US}\, D_c^{US}\, Post_t \\ &+ \delta_{CNUS}\, D_c^{US}\, d\tau_i^{CN-US} + \theta_{CNUS}\, Post_t\, d\tau_i^{CN-US} + \phi_{CNUS}\, D_c^{US}\, Post_t\, d\tau_i^{CN-US},\end{aligned}$$

and the triple interaction $D_c^{CN}\, Post_t\, d\tau_i^{US-CN}$, together with its lower-order terms that are not absorbed by the fixed effects, enters separately for products with high ($g = H$) and low ($g = L$) U.S. export exposure:[8]

$$\begin{aligned}\Gamma_{i,c,t}^{g} = \lambda_3^g\, d\tau_i^{US-CN} &+ \lambda_4^g\, D_c^{CN}\, Post_t + \lambda_5^g\, D_c^{CN}\, d\tau_i^{US-CN} \\ &+ \theta_{USCN}^g\, Post_t\, d\tau_i^{US-CN} + \phi_{USCN}^g\, D_c^{CN}\, Post_t\, d\tau_i^{US-CN}, \qquad g \in \{H, L\}.\end{aligned}$$

[7] Based on this definition, 10 HS2 industries are classified to have high U.S. export exposure: live animals (01), manufactures of straw (46), printed books and manuscripts (49), silk (50), other vegetable textile fibers (53), lead (78), vehicles other than railway (87), aircraft and spacecraft (88), musical instruments (92), and works of art and antiques (97).

[8] $Post_t$ and D_c^{US} do not enter $\Theta_{i,c,t}^{US}$ separately because they are absorbed by the HS4 $\times$ year and HS4 $\times$ country fixed effects, $\alpha_{s(i),t}$ and $\alpha_{s(i),c}$. Because $S_i^{US,H}$ is defined at the HS2 level, the same fixed effects also absorb $S_i^{US,H}$ and the products of D_c^{CN} and $Post_t$ with $S_i^{US,H}$ and $S_i^{US,L}$, which are therefore omitted from $\Gamma_{i,c,t}^{g}$.

The effect of the U.S. tariff on Korean imports from China is thus ϕ^{L}_{USCN} for products with $S_i^{US,L} = 1$ and ϕ^{H}_{USCN} for products with $S_i^{US,H} = 1$. Table 6 reports the estimates. We find that the response of Korean imports from China to the U.S. tariff varies sharply with Korean exporters' pre-shock exposure to the U.S. market.

The aggregate decline in Korean imports from China is driven entirely by products in which Korea has low pre-shock export exposure to the U.S. For products with $S_i^{US,L} = 1$ (HS10 products outside the top decile of Korean export share to the U.S. at the HS2 level), the estimated effect of the U.S. tariff on Korean imports from China is $\hat{\phi}^{L}_{USCN} = -2.38$, which mirrors the headline result in Table 2. For products with $S_i^{US,H} = 1$ (the top decile of Korean export share to the U.S.), the effect reverses sign: the estimated effect is $\hat{\phi}^{H}_{USCN} = 1.55$, which is positive and statistically significant at the 5% level. That is, for products in the top decile of Korean export share to the U.S., a 10 percentage-point increase in the U.S. tariff on a Chinese product leads to roughly a 15% increase in Korean imports of that product from China after 2017. In other words, Korean imports from China rise with the U.S. tariff for products in which Korean exporters were already heavily exposed to the U.S. market before the trade war. Using the pre-shock U.S.-export share as our proxy for average rerouting costs, the decline in Korean imports from China for products with low U.S.-export exposure and the increase for products with high U.S.-export exposure echo Prediction 3.

If the rise in Korean imports from China among products with high U.S. export exposure reflects rerouting of Chinese goods toward the U.S. market, the same products should show a corresponding rise in Korean exports to the U.S. To test this prediction, we estimate the export-side analogue of Equation (44) mirrored to the U.S. destination market, on Korean customs export data at the HS10 product $\times$ destination country $\times$ year level over the same period:

$$x_{i,c,t} \;=\; \Xi^{CN}_{i,c,t} + \Lambda^{H}_{i,c,t}\, S_i^{US,H} + \Lambda^{L}_{i,c,t}\, S_i^{US,L} + \alpha_{s(i),c} + \alpha_{s(i),t} + \varepsilon_{i,c,t}. \tag{45}$$

TABLE 6: Heterogeneity by Pre-shock U.S.-Export Exposure

	Dependent variable: Import Quantity
	(1)
CN × Post × $\tau^{\text{US-CN}} \times S_i^{\text{US,H}}$	1.545**
	(0.697)
Post × $\tau^{\text{US-CN}} \times S_i^{\text{US,H}}$	-1.044***
	(0.288)
CN × Post × $\tau^{\text{US-CN}} \times S_i^{\text{US,L}}$	-2.383***
	(0.893)
Post × $\tau^{\text{US-CN}} \times S_i^{\text{US,L}}$	0.452
	(0.932)
US × Post × $\tau^{\text{CN-US}}$	-4.125
	(2.829)
US × Post	1.321*
	(0.747)
Difference (High minus Low)	3.928***
	(1.117)
Fixed Effects	HS4 x Year, HS4 x Country
Observations	658,192

Notes: Each observation is at the HS10 product × origin country × year level. We include HS4 × year and HS4 × country fixed effects. Standard errors clustered at HS4 in parentheses. $S_i^{\text{US,H}}$ ($S_i^{\text{US,L}}$) is equal to one if HS10 product i falls (does not fall) in the top decile of the Korean export share to the U.S. at the HS2 level in 2017. Every term involving $\tau^{\text{US-CN}}$ or CN × Post is interacted separately with $S_i^{\text{US,H}}$ and $S_i^{\text{US,L}}$, so the coefficients on CN × Post × $\tau^{\text{US-CN}} \times S_i^{\text{US,H}}$ and CN × Post × $\tau^{\text{US-CN}} \times S_i^{\text{US,L}}$ are the effects for each group directly; the last row reports their difference. Other regressors are suppressed for brevity. * p<0.1, ** p<0.05, *** p<0.01.

where the outcome $x_{i,c,t}$ is the log export quantity of HS10 product i shipped to destination country c in year t, and D^{US} and D^{CN} are now destination-country dummies for the U.S. and China, respectively. The term $\Xi_{i,c,t}^{CN}$ collects the terms that involve China's retaliatory tariff, $d\tau_i^{CN-US}$, or the China-destination dummy, D_c^{CN}, which are common to products inside and outside the top decile of Korean export share to the U.S.[9]:

$$\Xi_{i,c,t}^{CN} = \alpha + \tilde{\beta}_2\, d\tau_i^{CN-US} + \tilde{\gamma}_{CN}\, D_c^{CN}\, Post_t + \tilde{\delta}_{CNUS}\, D_c^{CN}\, d\tau_i^{CN-US} + \tilde{\theta}_{CNUS}\, Post_t\, d\tau_i^{CN-US} + \psi_{CNUS}\, D_c^{CN}\, Post_t\, d\tau_i^{CN-US},$$

and the triple interaction $D_c^{US}\, Post_t\, d\tau_i^{US-CN}$, together with its lower-order terms that are

[9] As in Equation (44), $Post_t$, D_c^{CN}, D_c^{US}, $S_i^{US,H}$, and the products of D_c^{US} and $Post_t$ with $S_i^{US,H}$ and $S_i^{US,L}$ are absorbed by the fixed effects and therefore omitted.

not absorbed by the fixed effects, enters separately for the two groups:

$$\Lambda^{g}_{i,c,t} = \rho^{g}_{3}\, d\tau_i^{US-CN} + \rho^{g}_{4}\, D_c^{US}\, Post_t + \rho^{g}_{5}\, D_c^{US}\, d\tau_i^{US-CN} + \tilde{\theta}^{g}_{USCN}\, Post_t\, d\tau_i^{US-CN} + \psi^{g}_{USCN}\, D_c^{US}\, Post_t\, d\tau_i^{US-CN}, \qquad g \in \{H, L\},$$

Each tariff is paired with the destination market it affects: $d\tau_i^{US-CN}$ is interacted with the U.S. destination dummy D_c^{US}, while $d\tau_i^{CN-US}$ is interacted with the China destination dummy D_c^{CN}. The coefficients ψ^{H}_{USCN} and ψ^{L}_{USCN} measure the response of Korean exports to the U.S. for products inside and outside the top decile of Korean export share to the U.S., respectively.

We then examine whether products in the top decile of Korean export share to the U.S. also show a corresponding increase in Korean exports to the U.S. Table 7 reports estimates of Equation (45) with log export weight as the outcome variable. For products with $S_i^{US,L} = 1$, the estimated effect of the U.S. tariff on Korean exports to the U.S. is $\hat{\psi}^{L}_{USCN} = -2.92$, statistically significant at the 1% level: for products with low U.S. export exposure, Korean exports to the U.S. declined in the products subject to the U.S. tariff on China. For products with $S_i^{US,H} = 1$, the effect reverses sign: the estimated effect is $\hat{\psi}^{H}_{USCN} = 1.93$, which is positive and statistically significant at the 1% level. That is, for products in the top decile of Korean export share to the U.S., a 10 percentage-point increase in the U.S. tariff on a Chinese product leads to roughly a 19% increase in Korean exports of that product to the U.S. after 2017.[10] The response of Korean exports to China to China's retaliatory tariff, $\hat{\psi}_{CNUS}$, is *not statistically different from zero*, consistent with the lack of a systematic relationship between Korean imports from U.S. and $d\tau_i^{CN-US}$ noted in Section 2.3. The increase in Korean exports to the U.S. among products with high U.S.-export exposure, alongside the increase in Korean imports from China for those same products, echoes Prediction 4. In the model, these are the products where lower rerouting

[10]The estimated effect for products with $S_i^{US,H} = 1$ remains positive and statistically significant at least at the 5% level across all nine fixed-effects specifications we consider, ranging from 1.28 to 2.02.

TABLE 7: Export-Side DID: Rerouting to the U.S. Market

	Dependent variable: Export Quantity
	(1)
US × Post × $\tau^{\text{US-CN}} \times S_i^{\text{US,H}}$	1.934***
	(0.740)
Post × $\tau^{\text{US-CN}} \times S_i^{\text{US,H}}$	-0.497
	(0.932)
US × Post × $\tau^{\text{US-CN}} \times S_i^{\text{US,L}}$	-2.924***
	(0.528)
Post × $\tau^{\text{US-CN}} \times S_i^{\text{US,L}}$	0.034
	(0.590)
CN × Post × $\tau^{\text{CN-US}}$	0.236
	(0.991)
CN × Post	0.153
	(0.179)
Difference (High minus Low)	4.858***
	(0.888)
Fixed Effects	HS4 x Year, HS4 x Country
Observations	879,897

Notes: Each observation is at the HS10 product × destination country × year level, using Korean customs export data. We include HS4 × year and HS4 × country fixed effects. Standard errors clustered at HS4 in parentheses. $S_i^{\text{US,H}}$ ($S_i^{\text{US,L}}$) is equal to one if HS10 product i falls (does not fall) in the top decile of the Korean export share to the U.S. at the HS2 level in 2017. Every term involving $\tau^{\text{US-CN}}$ or US × Post is interacted separately with $S_i^{\text{US,H}}$ and $S_i^{\text{US,L}}$, so the coefficients on US × Post × $\tau^{\text{US-CN}} \times S_i^{\text{US,H}}$ and US × Post × $\tau^{\text{US-CN}} \times S_i^{\text{US,L}}$ are the effects for each group directly; the last row reports their difference. Other regressors are suppressed for brevity. * p<0.1, ** p<0.05, *** p<0.01.

costs make the increase in Chinese goods rerouted through Korea to the U.S. large enough to dominate the contraction in Korean-produced exports.

The Korean evidence therefore shows both decoupling and rerouting with China, conditional on the bystander country's pre-existing U.S.-export exposure. Korean imports from China decoupled in the bulk of HS lines with low U.S.-export exposure. Rerouting is concentrated in products with high U.S. export exposure, where Korean imports from China and Korean exports to the U.S. rise together with the U.S. tariff: goods flow in from China and out to the U.S. in the same product lines. Section 2 formalizes this dichotomy in a simple model.

4 Concluding Remarks

This paper studies how the 2018–19 U.S. tariffs on Chinese goods affected Korea's trade with China and the United States. In contrast to prior work emphasizing trade diversion and rerouting through countries not directly involved in the trade war, we find that Korea's imports from China declined while the prices Korean importers paid for Chinese goods increased.

We explain these findings using a simple analytical model of a global supply chain linking China, Korea, and the United States. The model features two channels. First, U.S. tariffs reduce Chinese sales to the U.S. and contract Chinese production scale. Under increasing returns to scale, this contraction raises Chinese producers' marginal costs and the prices faced by Korean importers, reducing Korea's imports from China and, through imported-input linkages, its exports to the U.S. Second, the tariffs make rerouting through Korea more attractive to Chinese exporters, increasing both Korean imports from China and Korean exports to the U.S. The relative importance of the two channels depends on Korea's pre-shock U.S.-export exposure, which affects the ease of rerouting through Korea.

Product-level Korean customs data support both channels. U.S. tariffs reduced Korean imports from China and raised the unit values of imports from China, with the import decline concentrated in industries exhibiting increasing returns to scale. In industries with high pre-shock U.S.-export exposure, however, Korean imports from China and exports to the U.S. both increased, consistent with rerouting.

In sum, the results show how Korea's position between Chinese producers and U.S. consumers shaped its response to the trade war. Dependence on Chinese inputs transmitted the contraction in Chinese production to Korean import prices, whereas Korea's established links to the U.S. market facilitated rerouting. The balance between these two linkages explains why the same tariff shock reduced Korea's imports from China on average but increased Korean imports from China and exports to the U.S. in industries with

high U.S.-export exposure. More broadly, our findings show that bilateral tariffs can reshape third-country links with the targeted economy differently across sectors, strengthening them through rerouting or trade diversion in some industries while weakening input-sourcing links in others.

References

Ahn, JaeBin, Amit K. Khandelwal, and Shang-Jin Wei, "The Role of Intermediaries in Facilitating Trade," *Journal of International Economics*, 2011, *84* (1), 73–85.

Amiti, Mary, Stephen J Redding, and David E Weinstein, "The impact of the 2018 tariffs on prices and welfare," *Journal of Economic Perspectives*, 2019, *33* (4), 187–210.

_ , _ , and _ , "Who's paying for the US tariffs? A longer-term perspective," *AEA Papers and Proceedings*, 2020, *110*, 541–546.

Antràs, Pol, Adrian Kulesza, and Andrea F Presbitero, "The Remains of the Trade: The U.S.–China Trade War and its Aftermath," Working Paper 35785, National Bureau of Economic Research September 2026.

Antweiler, Werner and Daniel Trefler, "Increasing returns and all that: a view from trade," *American Economic Review*, 2002, *92* (1), 93–119.

Benguria, Felipe and Felipe Saffie, "Escaping the Trade War: Finance and Relational Supply Chains in the Adjustment to Trade Policy Shocks," *Journal of International Economics*, 2024, *152*, 103987.

Bown, Chad P., "The US-China Trade War and Phase One Agreement," Peterson Institute for International Economics Working Paper 21-2 2021. Trade-war tariff timeline.

_ , "The US–China trade war and Phase One agreement," *Journal of Policy Modeling*, 2021, *43* (4), 805–843.

Cavalcanti, Tiago, Pedro Ogeda, and Emanuel Ornelas, "The US-China trade war creates jobs (elsewhere)," *Journal of International Economics*, 2026, *161*, 104249.

Cavallo, Alberto, Gita Gopinath, Brent Neiman, and Jenny Tang, "Tariff pass-through at the border and at the store: Evidence from us trade policy," *American Economic Review: Insights*, 2021, *3* (1), 19–34.

Conteduca, Francesco Paolo, Marco Errico, Fabrizio Leone, Ludovic Panon, and Giacomo Romanini, "The impact of trade wars on firms in third countries," Technical Report 12629, CESifo 2026.

Fajgelbaum, Pablo D. and Amit K. Khandelwal, "The Economic Impacts of the US-China Trade War," *Annual Review of Economics*, 2022, *14*, 205–228.

_ , Pinelopi K. Goldberg, Patrick J. Kennedy, and Amit K. Khandelwal, "The Return to Protectionism," *Quarterly Journal of Economics*, 2020, *135* (1), 1–55.

Fajgelbaum, Pablo, Pinelopi Goldberg, Patrick Kennedy, Amit Khandelwal, and Daria Taglioni, "The US-China trade war and global reallocations," *American Economic Review: Insights*, 2024, *6* (2), 295–312.

Flaaen, Aaron, Ali Hortaçsu, and Felix Tintelnot, "The production relocation and price effects of US trade policy: The case of washing machines," *American Economic Review*, 2020, *110* (7), 2103–2127.

_ and Justin R Pierce, "Disentangling the effects of the 2018-2019 tariffs on a globally connected U.S. manufacturing sector," *The Review of Economics and Statistics*, 2024, pp. 1–45.

Freund, Caroline, Aaditya Mattoo, Alen Mulabdic, and Michele Ruta, "Is US trade policy reshaping global supply chains?," *Journal of International Economics*, 2024, *152*, 104011.

Ganapati, Sharat, Woan Foong Wong, and Oren Ziv, "Entrepôt: Hubs, Scale, and Trade Costs," *American Economic Journal: Macroeconomics*, 2024, *16* (4), 239–278.

Garred, Jason and Song Yuan, "Relocation from China (with Chinese characteristics)," *Journal of Development Economics*, 2025, *176*, 103510.

Grossman, Gene M, Elhanan Helpman, and Stephen J Redding, "When tariffs disrupt global supply chains," *American Economic Review*, 2024, *114* (4), 988–1029.

Hardy, Bryan and Felipe Saffie, "From Carry Trades to Trade Credit: Financial Intermediation by Non-Financial Corporations," *Journal of International Economics*, 2024, *152*, 103988.

Iyoha, Ebehi, Edmund J. Malesky, Jaya Wen, Sung-Ju Wu, and Bo Feng, "Exports in Disguise: Trade Rerouting During the US-China Trade War," 2024. Working paper.

Jiang, Lingduo, Yi Lu, Hong Song, and Guofeng Zhang, "Responses of Exporters to Trade Protectionism: Inferences from the US-China Trade War," *Journal of International Economics*, 2023, *140*, 103687.

Jiao, Yang, Zhikuo Liu, Zhiwei Tian, and Xiaxin Wang, "The impacts of the U.S. trade war on Chinese exporters," *Review of Economics and Statistics*, 2024, *106* (6), 1576–1587.

Kucheryavyy, Konstantin, Gary Lyn, and Andrés Rodríguez-Clare, "Grounded by Gravity: A Well-Behaved Trade Model with Industry-Level Economies of Scale," *American Economic Journal: Macroeconomics*, April 2023, *15* (2), 372–412.

Ma, Hong, Jingxin Ning, and Yuan Xu, "Reshaping factory Asia: How did the US-China trade war affect the Asian-Pacific supply chain?," *Journal of International Economics*, 2026, p. 104327.

Mayr-Dorn, Karin, Gaia Narciso, Duc Anh Dang, and Hien Phan, "Trade diversion and labor market adjustment: Vietnam and the US-China trade war," *European Economic Review*, 2026, p. 105282.

Shen, Binchao, "Have global value chains shifted?," *Journal of International Economics*, 2026, *161*, 104264.

Utar, Hâle, Alfonso Cebreros Zurita, and Luis Bernardo Torres Ruiz, "The US-China Trade War and the Relocation of Global Value Chains to Mexico," *Review of Economics and Statistics*, 2025, pp. 1–47.

Utar, Hale, Luhang Wang, and Xiaodong Zhu, "Trade Diversion and Labor Market Outcomes," 2025. Working paper.

A Proofs

A.1 Proof of Lemma 1

The routing rule (10) and Assumption 1 give

$$n_i = \Pr\left(\varphi_{ij} < \frac{\tau}{1+\tau}\right) = 1 - \left(1 - \frac{\tau}{1+\tau}\right)^{\kappa_i} = 1 - (1+\tau)^{-\kappa_i}. \tag{46}$$

The continuous distribution assigns zero mass to the cutoff. For $\tau > 0$, rearranging gives

$$\kappa_i = -\frac{\log(1-n_i)}{\log(1+\tau)}, \tag{47}$$

For fixed $\tau > 0$, each $\kappa_i \in (0, \infty)$ determines a unique rerouting share $n_i \in (0,1)$, increasing in κ_i. Holding the other argument fixed, partial differentiation gives

$$\frac{dn_i}{d\tau} = \kappa_i(1+\tau)^{-\kappa_i - 1} = \frac{\kappa_i}{1+\tau}(1-n_i), \tag{48}$$

$$\frac{\partial n_i}{\partial \kappa_i} = \log(1+\tau)(1+\tau)^{-\kappa_i}, \tag{49}$$

$$\kappa_i \frac{\partial n_i}{\partial \kappa_i} = -(1-n_i)\log(1-n_i). \tag{50}$$

At $\tau = 0$, the right-hand tariff derivative is $dn_i/d\tau|_{0+} = \kappa_i$.

A.2 Proof of Lemma 2

Combining (21), (9), and (26) gives

$$P_{iU} = (\mu w^C)^{1/(1-\alpha\sigma)} \left[\frac{\kappa_i + (\eta-1)(1+\tau)^{-(\kappa_i+\eta-1)}}{\kappa_i + \eta - 1}\right]^{-1/[(\eta-1)(1-\alpha\sigma)]}. \tag{51}$$

This gives a unique positive U.S. price index under the maintained assumptions. Chinese output and supply prices then satisfy

$$Y_i^C = (\mu w^C)^{-\sigma/(1-\alpha\sigma)} \left[\frac{\kappa_i + (\eta-1)(1+\tau)^{-(\kappa_i+\eta-1)}}{\kappa_i+\eta-1} \right]^{\sigma/[(\eta-1)(1-\alpha\sigma)]}, \tag{52}$$

$$p_i^{CD} = p_{iK}^C = \mu w^C P_{iU}^{\alpha\sigma}. \tag{53}$$

Equations (14) and (15) determine p_i^K uniquely, and (27) then determines Y_i^K. Equations (24), (16), and (23) determine the remaining quantities.

A.3 Proof of Proposition 1

For fixed finite $\tau \geq 0$, letting $\kappa_i \downarrow 0$ in (51) and (53) gives the no-rerouting prices:

$$P_{iU} = \left[\mu w^C (1+\tau)\right]^{1/(1-\alpha\sigma)}, \qquad p_{iK}^C = \mu w^C P_{iU}^{\alpha\sigma}. \tag{54}$$

In this limit, (17) and (23) give $n_i = R_i = 0$. Taking logs and totally differentiating these prices, $Y_i^C = P_{iU}^{-\sigma}$, and $q_{iK}^C = (p_{iK}^C)^{-\sigma}$ with respect to τ proves Proposition 1.

For $\kappa_i > 0$, log differentiation of (51) with respect to τ gives

$$\frac{d\log P_{iU}}{d\tau} = \frac{(1+\tau)^{-(\kappa_i+\eta)}}{1-\alpha\sigma} \frac{\kappa_i+\eta-1}{\kappa_i+(\eta-1)(1+\tau)^{-(\kappa_i+\eta-1)}} > 0. \tag{55}$$

Equations (53) and (24) therefore imply

$$\frac{d\log p_{iK}^C}{d\tau} = \alpha\sigma \frac{d\log P_{iU}}{d\tau} > 0, \tag{56}$$

$$\frac{d\log q_{iK}^C}{d\tau} = -\alpha\sigma^2 \frac{d\log P_{iU}}{d\tau} < 0. \tag{57}$$

When $\alpha = 0$, neither p_{iK}^C nor q_{iK}^C responds to the tariff:

$$\frac{d \log p_{iK}^C}{d\tau} = 0, \qquad \frac{d \log q_{iK}^C}{d\tau} = 0.$$

The following lemma gives the remaining tariff responses.

Lemma 3 (Auxiliary equilibrium responses). *For every $\kappa_i > 0$, the remaining equilibrium responses used to prove Propositions 3 and 4 are*

$$\frac{d \log Y_i^C}{d\tau} = -\sigma \frac{d \log P_{iU}}{d\tau} < 0, \tag{58}$$

$$\frac{d \log p_{iU}^K}{d\tau} = \gamma\alpha\sigma \frac{d \log P_{iU}}{d\tau} > 0, \tag{59}$$

$$\frac{d \log Y_i^K}{d\tau} = (\eta - \sigma - \gamma\alpha\sigma\eta) \frac{d \log P_{iU}}{d\tau}, \tag{60}$$

$$\frac{d \log m_{iK}^C}{d\tau} = (\eta - \sigma - \gamma\alpha\sigma\eta - \alpha\sigma(1-\gamma)) \frac{d \log P_{iU}}{d\tau}. \tag{61}$$

If $\gamma\alpha\sigma\eta > \eta - \sigma$, both $dY_i^K/d\tau$ and $dm_{iK}^C/d\tau$ are negative.

Proof. Taking logs and totally differentiating (26), (53), (14), (15), and (27) with respect to τ gives the first three responses. Applying the same steps to (16) gives

$$\frac{d \log m_{iK}^C}{d\tau} = \frac{d \log Y_i^K}{d\tau} - (1-\gamma) \frac{d \log p_{iK}^C}{d\tau}, \tag{62}$$

which yields (61). Under $\gamma\alpha\sigma\eta > \eta - \sigma$,

$$\eta - \sigma - \gamma\alpha\sigma\eta < 0, \tag{63}$$

$$\eta - \sigma - \gamma\alpha\sigma\eta - \alpha\sigma(1-\gamma) < 0. \tag{64}$$

The Korean export price response is positive because $\gamma\alpha\sigma > 0$. □

A.4 Proof of Proposition 2

Without rerouting, $E_i = Y_i^K$ and $M_i = q_{iK}^C + m_{iK}^C$. Taking logs and totally differentiating (15), (14), and (27) with respect to τ, using Proposition 1, gives (33). Log differentiation of (16) gives

$$\begin{aligned} \frac{d\log m_{iK}^C}{d\tau} &= \frac{d\log Y_i^K}{d\tau} - (1-\gamma)\frac{d\log p_{iK}^C}{d\tau} \\ &= \frac{\eta - \sigma - \gamma\alpha\sigma\eta - \alpha\sigma(1-\gamma)}{(1-\alpha\sigma)(1+\tau)}. \end{aligned} \tag{65}$$

Under (34), Korean output, exports, and intermediate imports decline. Proposition 1 gives lower final imports, so total imports decline. More generally,

$$\frac{dM_i}{d\tau} = \frac{-\alpha\sigma^2 q_{iK}^C + [\eta - \sigma - \gamma\alpha\sigma\eta - \alpha\sigma(1-\gamma)]m_{iK}^C}{(1-\alpha\sigma)(1+\tau)}. \tag{66}$$

Total imports decline if $\gamma\alpha\sigma\eta + \alpha\sigma(1-\gamma) \geq \eta - \sigma$. If this condition fails, they still decline when the decline in final imports exceeds the increase in intermediate imports. Thus (34) is sufficient but not necessary. When $\alpha = 0$, Chinese supply prices and final imports are unchanged, while

$$\frac{dM_i}{d\tau} = \frac{(\eta-\sigma)m_{iK}^C}{1+\tau} > 0, \qquad \frac{dE_i}{d\tau} = \frac{(\eta-\sigma)E_i}{1+\tau} > 0. \tag{67}$$

A.5 Proof of Proposition 3

For $\tau > 0$, total differentiation of (29) with respect to τ, using Equation (55), Lemma 3, and (23), gives

$$\frac{dq_{iK}^C}{d\tau} = -\alpha\sigma^2 q_{iK}^C \frac{d\log P_{iU}}{d\tau}, \tag{68}$$

$$\frac{dm_{iK}^C}{d\tau} = -\left[\gamma\alpha\sigma\eta - (\eta-\sigma) + \alpha\sigma(1-\gamma)\right] m_{iK}^C \frac{d\log P_{iU}}{d\tau}, \tag{69}$$

$$\frac{dR_i}{d\tau} = \kappa_i(1+\tau)^{-(\kappa_i+\eta+1)}(p_i^{CD})^{-\eta} P_{iU}^{\eta-\sigma} - \left[\alpha\sigma\eta - (\eta-\sigma)\right] R_i \frac{d\log P_{iU}}{d\tau}, \tag{70}$$

$$\begin{aligned}\frac{dM_i}{d\tau} &= \kappa_i(1+\tau)^{-(\kappa_i+\eta+1)}(p_i^{CD})^{-\eta}P_{iU}^{\eta-\sigma} - \alpha\sigma^2 q_{iK}^C \frac{d\log P_{iU}}{d\tau} \\ &\quad - \left[\gamma\alpha\sigma\eta - (\eta-\sigma) + \alpha\sigma(1-\gamma)\right] m_{iK}^C \frac{d\log P_{iU}}{d\tau} \\ &\quad - \left[\alpha\sigma\eta - (\eta-\sigma)\right] R_i \frac{d\log P_{iU}}{d\tau}. \end{aligned} \tag{71}$$

The positive term in $dM_i/d\tau$ comes from firms that begin rerouting. The negative terms reflect lower imports for final consumption, intermediate imports, and shipments by firms already rerouting. These terms are negative because $\gamma\alpha\sigma\eta > \eta - \sigma$, $\gamma \in (0,1)$, and $d\log P_{iU}/d\tau > 0$.

The ratios of the positive term to the magnitudes of the three negative terms are

$$\begin{aligned}&\frac{\kappa_i(1+\tau)^{-(\kappa_i+\eta+1)}(p_i^{CD})^{-\eta}P_{iU}^{\eta-\sigma}}{\alpha\sigma^2 q_{iK}^C \, d\log P_{iU}/d\tau} \\ &\quad = \frac{(1-\alpha\sigma)\kappa_i}{\alpha\sigma^2(1+\tau)}\left[\frac{\kappa_i + (\eta-1)(1+\tau)^{-(\kappa_i+\eta-1)}}{\kappa_i+\eta-1}\right]^{(\sigma-1)/(\eta-1)}, \end{aligned} \tag{72}$$

$$\begin{aligned}&\frac{\kappa_i(1+\tau)^{-(\kappa_i+\eta+1)}(p_i^{CD})^{-\eta}P_{iU}^{\eta-\sigma}}{\left[\gamma\alpha\sigma\eta - (\eta-\sigma) + \alpha\sigma(1-\gamma)\right] m_{iK}^C \, d\log P_{iU}/d\tau} \\ &\quad = \frac{\mu}{\gamma}\left[\mu(w^K)^{1-\gamma}(\mu w^C)^{(\gamma-1)/(1-\alpha\sigma)}\right]^{\eta-1} \frac{(1-\alpha\sigma)\kappa_i}{\left[\gamma\alpha\sigma\eta - (\eta-\sigma) + \alpha\sigma(1-\gamma)\right](1+\tau)} \\ &\qquad \times \left[\frac{\kappa_i + (\eta-1)(1+\tau)^{-(\kappa_i+\eta-1)}}{\kappa_i+\eta-1}\right]^{1+\frac{\alpha\sigma(1-\gamma)}{1-\alpha\sigma}}, \end{aligned} \tag{73}$$

$$\begin{aligned}&\frac{\kappa_i(1+\tau)^{-(\kappa_i+\eta+1)}(p_i^{CD})^{-\eta}P_{iU}^{\eta-\sigma}}{\left[\alpha\sigma\eta - (\eta-\sigma)\right] R_i \, d\log P_{iU}/d\tau} \\ &\quad = \frac{1-\alpha\sigma}{\left[\alpha\sigma\eta - (\eta-\sigma)\right](1+\tau)} \frac{\dfrac{\kappa_i + (\eta-1)(1+\tau)^{-(\kappa_i+\eta-1)}}{\kappa_i+\eta-1}}{\displaystyle\int_1^{1+\tau} z^{-(\kappa_i+\eta+1)}dz}. \end{aligned} \tag{74}$$

Holding τ and other parameters fixed, the integral representation and partial derivatives with respect to κ_i are

$$\frac{\kappa_i + (\eta-1)(1+\tau)^{-(\kappa_i+\eta-1)}}{\kappa_i+\eta-1} = 1 - (\eta-1)\int_1^{1+\tau} z^{-(\kappa_i+\eta)}dz, \tag{75}$$

$$\frac{\partial}{\partial \kappa_i}\left[\frac{\kappa_i+(\eta-1)(1+\tau)^{-(\kappa_i+\eta-1)}}{\kappa_i+\eta-1}\right]=(\eta-1)\int_1^{1+\tau}(\log z)z^{-(\kappa_i+\eta)}dz>0, \tag{76}$$

$$\frac{\partial}{\partial \kappa_i}\int_1^{1+\tau} z^{-(\kappa_i+\eta+1)}dz=-\int_1^{1+\tau}(\log z)z^{-(\kappa_i+\eta+1)}dz<0. \tag{77}$$

The exponents on the common fraction in (72) and (73) are positive:

$$\frac{\sigma-1}{\eta-1}>0, \qquad 1+\frac{\alpha\sigma(1-\gamma)}{1-\alpha\sigma}>0. \tag{78}$$

Thus all three ratios in (72)–(74) strictly increase with κ_i. As $\kappa_i \downarrow 0$, the first two ratios converge to zero, while the ratio in (74) converges to

$$\frac{(1-\alpha\sigma)\eta}{[\alpha\sigma\eta-(\eta-\sigma)]\left[(1+\tau)^{\eta}-1\right]} \tag{79}$$

All three ratios diverge to infinity as $\kappa_i \to \infty$.

Dividing (71) by its positive rerouting term and using (72)–(74) gives

$$\begin{aligned}\frac{dM_i/d\tau}{\kappa_i(1+\tau)^{-(\kappa_i+\eta+1)}(p_i^{CD})^{-\eta}P_{iU}^{\eta-\sigma}} &= 1-\frac{\alpha\sigma^2(1+\tau)}{(1-\alpha\sigma)\kappa_i}\left[\frac{\kappa_i+(\eta-1)(1+\tau)^{-(\kappa_i+\eta-1)}}{\kappa_i+\eta-1}\right]^{-\frac{\sigma-1}{\eta-1}}\\ &-\frac{\gamma}{\mu}\left[\mu(w^K)^{1-\gamma}(\mu w^C)^{(\gamma-1)/(1-\alpha\sigma)}\right]^{1-\eta}\frac{[\gamma\alpha\sigma\eta-(\eta-\sigma)+\alpha\sigma(1-\gamma)](1+\tau)}{(1-\alpha\sigma)\kappa_i}\\ &\times\left[\frac{\kappa_i+(\eta-1)(1+\tau)^{-(\kappa_i+\eta-1)}}{\kappa_i+\eta-1}\right]^{-\frac{1-\gamma\alpha\sigma}{1-\alpha\sigma}}\\ &-\frac{[\alpha\sigma\eta-(\eta-\sigma)](1+\tau)}{1-\alpha\sigma}\frac{1-(1+\tau)^{-(\kappa_i+\eta)}}{\kappa_i+\eta}\frac{\kappa_i+\eta-1}{\kappa_i+(\eta-1)(1+\tau)^{-(\kappa_i+\eta-1)}}.\end{aligned} \tag{80}$$

The subtracted terms are the reciprocals of the ratios in (72)–(74). Hence the right-hand side of (80) is continuous and strictly increasing in κ_i, with limits of minus infinity as $\kappa_i \downarrow 0$ and one as $\kappa_i \to \infty$. Since the denominator is positive, $dM_i/d\tau$ has a unique zero $\kappa_i^M(\tau)$, is negative below it, and positive above it.

At $\tau = 0$, Equation (55) gives $d\log P_{iU}/d\tau = 1/(1-\alpha\sigma)$. Substitution into (71) gives

the right-hand tariff derivative:

$$\begin{aligned}\left.\frac{dM_i}{d\tau}\right|_{0+} &= \kappa_i \left.(p_i^{CD})^{-\eta} P_{iU}^{\eta-\sigma}\right|_{\tau=0} \\ &\quad - \frac{\alpha\sigma^2}{1-\alpha\sigma} \left. q_{iK}^C \right|_{\tau=0} \\ &\quad - \frac{\gamma\alpha\sigma\eta - (\eta-\sigma) + \alpha\sigma(1-\gamma)}{1-\alpha\sigma} \left. m_{iK}^C \right|_{\tau=0}. \end{aligned} \tag{81}$$

Therefore,

$$\kappa_i^M(0) = \frac{\alpha\sigma^2 \left. q_{iK}^C \right|_{\tau=0} + \left[\gamma\alpha\sigma\eta - (\eta-\sigma) + \alpha\sigma(1-\gamma)\right] \left. m_{iK}^C \right|_{\tau=0}}{(1-\alpha\sigma) \left.(p_i^{CD})^{-\eta} P_{iU}^{\eta-\sigma}\right|_{\tau=0}} > 0. \tag{82}$$

Prices and quantities at $\tau = 0$ are independent of κ_i, so the derivative in (81) is linear in κ_i, with a positive slope and a negative intercept.

A.6 Proof of Proposition 4

For $\tau > 0$, total differentiation with respect to τ gives

$$\frac{dY_i^K}{d\tau} = -\left[\gamma\alpha\sigma\eta - (\eta-\sigma)\right] Y_i^K \frac{d\log P_{iU}}{d\tau}, \tag{83}$$

$$\begin{aligned}\frac{dE_i}{d\tau} &= \kappa_i (1+\tau)^{-(\kappa_i+\eta+1)} (p_i^{CD})^{-\eta} P_{iU}^{\eta-\sigma} \\ &\quad - \left[\gamma\alpha\sigma\eta - (\eta-\sigma)\right] Y_i^K \frac{d\log P_{iU}}{d\tau} \\ &\quad - \left[\alpha\sigma\eta - (\eta-\sigma)\right] R_i \frac{d\log P_{iU}}{d\tau}. \end{aligned} \tag{84}$$

The positive term in $dE_i/d\tau$ comes from firms that begin rerouting. The negative terms reflect lower Korean-produced exports and shipments by firms already rerouting. The ratio of the positive term to the reduction in Korean-produced exports is

$$\frac{\kappa_i (1+\tau)^{-(\kappa_i+\eta+1)} (p_i^{CD})^{-\eta} P_{iU}^{\eta-\sigma}}{\left[\gamma\alpha\sigma\eta - (\eta-\sigma)\right] Y_i^K \, d\log P_{iU}/d\tau}$$

$$= \left[\mu(w^K)^{1-\gamma}(\mu w^C)^{(\gamma-1)/(1-\alpha\sigma)}\right]^{\eta} \frac{(1-\alpha\sigma)\kappa_i}{[\gamma\alpha\sigma\eta-(\eta-\sigma)]\,(1+\tau)}$$
$$\times \left[\frac{\kappa_i+(\eta-1)(1+\tau)^{-(\kappa_i+\eta-1)}}{\kappa_i+\eta-1}\right]^{1+\frac{\alpha\sigma\eta(1-\gamma)}{(\eta-1)(1-\alpha\sigma)}}. \tag{85}$$

The exponent on the last bracketed term is positive:

$$1+\frac{\alpha\sigma\eta(1-\gamma)}{(\eta-1)(1-\alpha\sigma)}>0. \tag{86}$$

For fixed τ and other parameters, the ratio in (85) strictly increases with κ_i, as in the proof of Proposition 3. It converges to zero as $\kappa_i \downarrow 0$ and diverges to infinity as $\kappa_i \to \infty$. The ratio of additional rerouted exports to the reduction in existing rerouted shipments is (74). It also increases strictly with κ_i and, as $\kappa_i \downarrow 0$, converges to

$$\frac{(1-\alpha\sigma)\eta}{[\alpha\sigma\eta-(\eta-\sigma)]\left[(1+\tau)^{\eta}-1\right]} \tag{87}$$

This ratio diverges to infinity as $\kappa_i \to \infty$.

Dividing (84) by its positive rerouting term and using (85) and (74) gives

$$\frac{dE_i/d\tau}{\kappa_i(1+\tau)^{-(\kappa_i+\eta+1)}(p_i^{CD})^{-\eta}P_{iU}^{\eta-\sigma}} = 1$$
$$-\left[\mu(w^K)^{1-\gamma}(\mu w^C)^{(\gamma-1)/(1-\alpha\sigma)}\right]^{-\eta}\frac{[\gamma\alpha\sigma\eta-(\eta-\sigma)](1+\tau)}{(1-\alpha\sigma)\kappa_i}$$
$$\times\left[\frac{\kappa_i+(\eta-1)(1+\tau)^{-(\kappa_i+\eta-1)}}{\kappa_i+\eta-1}\right]^{-1-\frac{\alpha\sigma\eta(1-\gamma)}{(\eta-1)(1-\alpha\sigma)}}$$
$$-\frac{[\alpha\sigma\eta-(\eta-\sigma)](1+\tau)}{1-\alpha\sigma}\frac{1-(1+\tau)^{-(\kappa_i+\eta)}}{\kappa_i+\eta}\frac{\kappa_i+\eta-1}{\kappa_i+(\eta-1)(1+\tau)^{-(\kappa_i+\eta-1)}}. \tag{88}$$

The subtracted terms are the reciprocals of the ratios in (85) and (74). Hence the right-hand side of (88) is continuous and strictly increasing in κ_i, with limits of minus infinity as $\kappa_i \downarrow 0$ and one as $\kappa_i \to \infty$. Since the denominator is positive, $dE_i/d\tau$ has a unique zero

$\kappa_i^E(\tau)$, is negative below it, and positive above it.

At $\tau = 0$, the right-hand tariff derivative obtained from (84) is

$$\left.\frac{dE_i}{d\tau}\right|_{0+} = \kappa_i \left.\left(p_i^{CD}\right)^{-\eta} P_{iU}^{\eta-\sigma}\right|_{\tau=0} - \frac{\gamma\alpha\sigma\eta - (\eta-\sigma)}{1-\alpha\sigma} \left.Y_i^K\right|_{\tau=0}. \tag{89}$$

Therefore,

$$\kappa_i^E(0) = \frac{\left[\gamma\alpha\sigma\eta - (\eta-\sigma)\right] \left.Y_i^K\right|_{\tau=0}}{(1-\alpha\sigma) \left.\left(p_i^{CD}\right)^{-\eta} P_{iU}^{\eta-\sigma}\right|_{\tau=0}} > 0. \tag{90}$$

Prices and quantities at $\tau = 0$ are independent of κ_i, so the derivative in (89) is linear in κ_i, with a positive slope and a negative intercept.

B Additional Results

TABLE 8: Baseline DID with Different Fixed Effects

	Dependent variable: Weight								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
CN × Post × $\tau^{\text{US-CN}}$	-1.648***	-1.475***	-1.341***	-1.667***	-1.491***	-1.354***	-2.373***	-2.644***	-2.166**
	(0.542)	(0.534)	(0.490)	(0.548)	(0.538)	(0.493)	(0.893)	(0.923)	(0.953)
US × Post × $\tau^{\text{CN-US}}$	-3.049	-2.918	-2.716	-3.049	-2.928	-2.727	-4.119	-4.757	-4.709
	(2.230)	(2.312)	(2.289)	(2.238)	(2.325)	(2.302)	(2.829)	(3.124)	(3.107)
US × Post	1.031*	0.951	0.948	1.033*	0.955	0.952	1.320*	1.403*	1.485*
	(0.593)	(0.621)	(0.618)	(0.596)	(0.625)	(0.623)	(0.748)	(0.845)	(0.845)
CN × Post	0.257**	0.216*	0.202*	0.262**	0.221*	0.206*	0.465**	0.532**	0.426*
	(0.128)	(0.120)	(0.109)	(0.129)	(0.121)	(0.110)	(0.233)	(0.247)	(0.247)
HS Codes	HS4	HS6	HS8	HS4	HS6	HS8	HS4	HS6	HS8
HS × Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE				Yes	Yes	Yes			
HS × Year FE							Yes	Yes	Yes
Observations	658,248	622,149	592,844	658,248	622,149	592,844	658,192	621,001	588,722

Notes: Each observation is at the HS10 product × origin country × year level. Column (7) reports our baseline results. Standard errors clustered at HS4 in parentheses. * p<0.1, ** p<0.05, *** p<0.01.